Forcing the Π^1_n-Uniformization Property

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Abstract

We generically construct a model in which the Π^1_3-uniformization property is true, thus lowering the best known consistency strength from the existence of $M_1^{\#}$ to just ZFC. The forcing construction also works over canonical inner models with Woodin cardinals, which yields, for the first time, universes where the Π^1_{2n}-uniformization property holds, thus producing models which contradict the natural PD-induced pattern.

1 Introduction

The question of finding nicely definable choice functions for a definable family of sets is an old and well-studied subject in descriptive set theory; see, for instance, [12, 17]. The uniformization problem, first mentioned by N. Lusin in 1930 (see [14]), asks to find choice functions which lie at the same projective level as the set they aim to uniformize. Recall that for $A \subseteq 2^\omega \times 2^\omega$, a *uniformization* of A is a function

$$f : \mathrm{pr}_1(A) \longrightarrow 2^\omega$$

such that

$$\mathrm{graph}(f) \subseteq A.$$

Equivalently, for every $x \in \mathrm{pr}_1(A)$ one has $(x, f(x)) \in A$.

Definition 1.1. *We say that a pointclass Γ has the uniformization property iff every element of Γ admits a uniformization in Γ.*

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It is a classical result due to M. Kondo [13] that lightface Π^1_1-sets do have the uniformization property, this also yields the uniformization property for Σ^1_2-sets. This is as much as ZFC can prove about uniformization. In the constructible universe L, as shown by J. Addison in [1], Σ^1_n does have the uniformization property for $n \geq 3$, which follows from the existence of a Σ^1_2-good wellorder of the reals; for background on projective wellorders see also [2, 4]. Thus the Π^1_n-uniformization fails for $n \geq 3$. On the other hand, by the celebrated results of Y. Moschovakis (see [16], Theorem 1), $\mathbf{\Delta^1_{2n}}$-projective determinacy implies Π^1_{2n+1}-uniformization, yet the determinacy assumption exceeds in logical strength ZFC. It is known due to H. W. Woodin (see [18],), that $\mathbf{\Delta^1_2}$-projective determinacy implies the existence of $M_1^{\#}$, hence yields an inner model with a Woodin cardinal. As with other regularity properties of the reals like Lebesgue measurability or Baire property, which both follow from PD as well, it is natural to ask whether the Π^1_3-uniformization property bears large cardinal strength as well. We shall answer it negatively.

Theorem. *There is a generic extension of L in which the Π^1_3-uniformization property is true.*

The same uniformization construction also applies over canonical inner models with Woodin cardinals. This can be used to obtain better lower bounds in terms of consistency strength for the Π^1_n uniformization property for odd n. For even n we can produce for the first time models where the Π^1_n uniformization property holds true.

Theorem. *Let M_n be the canonical inner model with n Woodin cardinals. Then there is a generic extension of M_n in which the Π^1_{n+3} uniformization property holds true.*

Since the first version of this paper was written, the method developed here has led to several further constructions. A variant combines Π-uniformization with a projective well-order of the reals under CH [7]; another combines global Σ-uniformization with a large continuum and a projective well-order [8]; and the Σ-side of the local Delfino problem combines related coding methods with projective regularity and uniformization [9]. These later papers should be viewed as refinements and applications of the forcing machinery introduced here. The present paper isolates the original fixed-point and allowability mechanism needed to force projective Π-uniformization without assuming the corresponding determinacy strength.

Questions concerning the forceability of (local) consequences of PD do have a long tradition in set theory. There is a vast body of literature concerning the forcability of local levels of the projective hierarchy satisfying (Boolean combinations of) the Baire property, the perfect set property or Lebesgue measurability; see for instance [5]. There has been very little progress in the past, however, concerning similar questions for the separation,

the reduction and the uniformization property. Indeed, even the question of whether one can force the Σ^1_3-separation property, which is the weakest of said properties, remained an open problem for 50 years (see [15], Problem 3029).

This article continues this line of research and develops the forcing construction used in the subsequent work mentioned above. It is organized as follows: in the preliminaries section, we briefly introduce the forcings which we will use in the proof and produce a generic extension W of L which will be a well-suited ground model for our needs. We then start to prove the theorems from above. The main idea is to turn the problem of finding a partial order $\mathbb{P}$ which forces the Π^1_3-property into a fixed point problem. We shall define a derivation operator which acts on a specific set of forcing iterations of length ω_1. This operator will be applied transfinitely often, and will produce better and better approximations to the set of forcings we actually want to use in the end. The process is shown to converge in that eventually a fixed point $\mathcal{P}$, i.e. a suitable set of forcings is reached. Forcings which belong to this fixed point $\mathcal{P}$ allow for a certain, seemingly self-referential definition of an ω_1-length iteration which forces the Π^1_3-uniformization property over W. We then follow up, to alter the said process such that it becomes applicable to the canonical inner models M_n with n Woodin cardinals.

There are some similarities to [6], in particular the two proofs rely on a similar ground model W, which is a generic extension of L, and use a similar coding method which relies on a suitably chosen ω_1-sequence of ω_1-Suslin trees. However a more straightforward application of the ideas of [6] will fail badly to produce a model of the Π^1_3-uniformization property. As a consequence, a solution has to necessarily introduce several new ideas in order to succeed. The presentation of those is the goal of this paper.

2 Preliminaries

2.1 Notation

The notation we use will be mostly standard, we hope. We write $\mathbb{P} = (\mathbb{P}_\alpha : \alpha < \gamma)$ for a forcing iteration of length γ with initial segments $\mathbb{P}_\alpha$. The α-th factor of the iteration will be denoted with $\mathbb{P}(\alpha)$. Note here that we drop the dot on $\mathbb{P}(\alpha)$, even though $\mathbb{P}(\alpha)$ is in fact a $\mathbb{P}_\alpha$-name of a partial order.

2.2 The forcings which are used

The forcing notions used in the construction are standard. The preparatory extension is obtained solely by adding branches through a fixed sequence of Suslin trees, and the later coding uses the ground-model version of the ω_1-Cohen forcing together with Jensen–Solovay almost disjoint coding [11].

Recall that a Suslin tree is a normal tree of height ω_1 with countable levels and without an uncountable chain or antichain. Whenever S is a Suslin tree, we also write S for the forcing whose conditions are the nodes of S, ordered by end-extension. Thus forcing with S adds a cofinal branch through S.

Definition 2.1. *Let $\vec{S} = \langle S_\xi \mid \xi < \kappa \rangle$ be a sequence of Suslin trees. We call $\vec{S}$* independent *if, for every nonempty finite set $e \subseteq \kappa$, the product*

$$\prod_{\xi \in e} S_\xi$$

is again a Suslin tree.

For $A \subseteq \kappa$ let

$$\mathrm{Br}_A(\vec{S}) = \prod_{\xi \in A}^{\mathrm{fs}} S_\xi$$

denote the finite-support product of the corresponding branch forcings. When the sequence is fixed, we simply write Br_A.

Lemma 2.2. *Let $\vec{S} = \langle S_\xi \mid \xi < \kappa \rangle$ be an independent sequence of Suslin trees, where $\kappa \leq \omega_1$.*

1. *For every $A \subseteq \kappa$, the forcing Br_A is c.c.c.*
2. *If $A \subseteq B \subseteq \kappa$, then Br_A is a complete subforcing of Br_B, and*

$$\mathrm{Br}_B \cong \mathrm{Br}_A \times \mathrm{Br}_{B \setminus A}\,.$$

3. *If $\xi \notin A$, then*

$$\mathrm{Br}_A \Vdash \text{“}S_\xi \text{ is a Suslin tree.”}$$

Proof. For (1), let $\{p_\alpha \mid \alpha < \omega_1\}$ be a family of conditions in Br_A. After thinning it out, the supports form a Δ-system with finite root r, and the restrictions $p_\alpha \restriction r$ belong to the finite product $\prod_{\xi \in r} S_\xi$. By independence this finite product is Suslin, hence c.c.c., so two of the root restrictions are compatible. Since the remaining parts of their supports are disjoint, the corresponding two conditions in Br_A are compatible.

The factorization in (2) is the usual coordinatewise factorization of a finite-support product. For (3), observe that

$$\mathrm{Br}_A \times S_\xi \cong \mathrm{Br}_{A \cup \{\xi\}},$$

which is c.c.c. by (1). The standard product characterization of preservation of a Suslin tree now yields the assertion. □

We next recall the almost disjoint coding forcing due to Jensen and Solovay [11]. We identify subsets of ω with their characteristic functions. Let $D = \langle d_\alpha \mid \alpha < \omega_1 \rangle$ be an almost disjoint family of infinite subsets of ω, and let $X \subseteq \omega_1$.

Definition 2.3. *The almost disjoint coding forcing $\mathbb{A}_D(X)$ consists of pairs $(r, R) \in [\omega]^{<\omega} \times [D]^{<\omega}$. We put $(s, S) \leq (r, R)$ if and only if*

1. *$r \subseteq s$ and $R \subseteq S$;*
2. *whenever $\alpha \in X$ and $d_\alpha \in R$, then $s \cap d_\alpha = r \cap d_\alpha$.*

If $G \subseteq \mathbb{A}_D(X)$ is generic and $x_G = \bigcup\{r : (r, R) \in G\}$, then

$$\alpha \in X \quad \Longleftrightarrow \quad x_G \cap d_\alpha \text{ is finite.}$$

Fix a recursive injection $\# : 2^{<\omega} \to \omega$. If $\langle x_\xi : \xi < \omega_1^L \rangle$ enumerates the constructible reals in the canonical order, use $d_\xi = \{\#(x_\xi \restriction k) : k < \omega\}$ and $D = \langle d_\xi : \xi < \omega_1^L \rangle$.

Lemma 2.4. *For every $X \subseteq \omega_1$, the forcing $\mathbb{A}_D(X)$ is Knaster. In particular it is c.c.c. and preserves every ground-model Suslin tree.*

Proof. The Knaster property follows from the usual Δ-system argument applied to the finite side conditions. If T is a Suslin tree, then the product of a Knaster forcing with the c.c.c. forcing T is c.c.c. Hence $\mathbb{A}_D(X) \times T$ is c.c.c., which is equivalent to preservation of T as a Suslin tree. □

Finally, we write

$$\mathbb{C}_L(\omega_1) = ({}^{<\omega_1}2)^L$$

for the L-computed ω_1-Cohen forcing, ordered by end-extension. Thus conditions are countable binary sequences belonging to L. Whenever this forcing is used in an outer model of L, the notation refers to this fixed L-poset rather than to a recomputation in the ambient universe.

2.3 The ground model W of the iteration

We now define the ground model over which the main iteration will be carried out. We start with the canonical independent Suslin sequence already present in L and add a cofinal generic branch through every member of that sequence.

The following lemma gives the simultaneous form of Jensen's diamond construction which we use throughout the paper; see also [10]. We use the levelwise product of finitely many trees. This is a dense suborder of the corresponding finite forcing product.

Lemma 2.5. *There is a canonical L-definable sequence*

$$\vec{S} = \langle S_\xi \mid \xi < \omega_1 \rangle$$

of independent normal Suslin trees. The construction is given by a recursion which is uniform over L_{ω_1}.

Proof. Work in L. Fix the canonical L-diamond sequence $\langle a_\alpha \mid \alpha < \omega_1 \rangle$ and the canonical L-coding of finite sets of ordinals, countable trees, and subsets of their finite products. We recursively construct coherent countable approximations $\vec{S}^{\,\alpha} = \langle S^\alpha_\xi \mid \xi < \alpha \rangle$ such that each S^α_ξ is a normal splitting tree of height α and $S^\beta_\xi = S^\alpha_\xi \restriction \beta$ whenever $\xi < \beta < \alpha$. At limit stages we take unions.

It remains to describe the passage from α to $\alpha + 1$. For old coordinates $\xi < \alpha$, successor stages are handled by putting countably many immediate successors above each node on the top level. Suppose now that α is a nonzero limit. If a_α does not decode as a pair (e, A), where $e \in [\alpha]^{<\omega} \setminus \{\varnothing\}$ and A is a maximal antichain in the product

$$P^\alpha_e = \bigcup_{\delta<\alpha} \prod_{\xi \in e} (S^\alpha_\xi)_\delta,$$

then we use the $<_L$-least countable family of cofinal branches through each old tree which covers all old nodes, and put one new top node above each of these branches.

If a_α does decode such a pair (e, A), we choose the branch families for the coordinates in e simultaneously so as to seal A. Concretely, by a countable genericity argument in $\prod^{\mathrm{fs}}_{(\xi,n)\in e\times\omega} S^\alpha_\xi$, choose cofinal branches $b_{\xi,n}$, $\xi \in e$, $n < \omega$, such that they cover all old nodes and such that, for every $\nu : e \to \omega$, the tuple of new top nodes placed above $\langle b_{\xi,\nu(\xi)} \mid \xi \in e \rangle$ extends some member of A. The required dense sets are just the usual ones ensuring cofinality, coverage of old nodes, and the displayed sealing requirement; density of the last kind uses maximality of A in P^α_e. For $\xi \notin e$ we again use the default $<_L$-least covering family. Finally, the new coordinate α is initialized as the canonical finite-support binary tree of height $\alpha + 1$. This completes the recursion. Put $S_\xi = \bigcup_{\xi<\alpha<\omega_1} S^\alpha_\xi$. Then each S_ξ is a normal splitting tree of height ω_1 with countable levels.

We show that the sequence is independent. Fix a finite nonempty $e \subseteq \omega_1$ and let

$$P_e = \bigcup_{\alpha<\omega_1} \prod_{\xi \in e} (S_\xi)_\alpha$$

be the product. This is dense in the usual finite product of the corresponding tree forcings. Let $A \subseteq P_e$ be a maximal antichain in L. There is a club of $\alpha > \sup(e)$ such that $A \cap (P_e \restriction \alpha)$ is maximal in $P_e \restriction \alpha$, and, by coherence of the coding, a club on which the fixed code $x_{e,A}$ for (e, A) restricts to a code

for $(e, A \cap (P_e \restriction \alpha))$. By $\diamondsuit$, choose a limit α in the intersection of these clubs with $a_\alpha = x_{e,A} \cap \alpha$. At stage α the construction seals $A \cap (P_e \restriction \alpha)$. Hence every node on level α, and therefore every later node, extends a member of this initial antichain. Since A itself is an antichain, $A = A \cap (P_e \restriction \alpha)$, so A is countable. Thus P_e has no uncountable antichain. A cofinal branch through P_e would, by normal splitting, yield an uncountable antichain by choosing at successor levels nodes incompatible with the branch. Hence P_e is Suslin. Since e was arbitrary, $\vec{S}$ is independent. The construction is canonical from $<_L$, the diamond sequence, and the fixed coding, and is therefore uniform over L_{ω_1}. □

Lemma 2.6. *Let M be a transitive uncountable model of*

$$\mathsf{ZFC}^- + \text{“}\aleph_1 \text{ exists”}$$

such that

$$\omega_1^M = \omega_1^L = \omega_1.$$

There is a Σ_1-formula

$$\vartheta_{\mathrm{Sus}}(t, \gamma, \eta, \omega_1),$$

independent of M, which defines over M the relation

$$\{(t, \gamma, \eta) : \gamma, \eta < \omega_1 \text{ and } t \in (S_\eta)_\gamma\}.$$

Equivalently, there is a formula $\mathrm{CanSuslin}_L(X, \omega_1)$, which may be taken to be $\Sigma_1(\omega_1)$ over $H(\omega_2)$, such that in every ω_1-preserving outer model U of L,

$$U \models \mathrm{CanSuslin}_L(X, \omega_1) \quad \Longleftrightarrow \quad X = \vec{S}.$$

In particular, the canonical sequence $\vec{S}$ is computed in the same way in all the outer models considered below.

Proof. By Lemma 2.5, the sequence $\vec{S}$ is obtained by a canonical recursion over L_{ω_1}, and the corresponding level relation is $\Sigma_1(\omega_1)$-definable over L_{ω_1}. If M is transitive, satisfies ZFC^- + “$\aleph_1$ exists”, and has $\omega_1^M = \omega_1^L$, then M computes L_{ω_1} correctly. Hence M computes the same canonical diamond sequence, the same coding apparatus, and therefore the same recursive construction of $\vec{S}$. The same Σ_1-formula $\vartheta_{\mathrm{Sus}}(t, \gamma, \eta, \omega_1)$ consequently defines over M the relation $t \in (S_\eta)_\gamma$.

The formulation by $\mathrm{CanSuslin}_L(X, \omega_1)$ is obtained by saying that X is the union of the unique system produced by this canonical recursion. Since every ω_1-preserving outer model of L has the same L_{ω_1}, it satisfies this formula exactly for the true sequence $\vec{S}$. □

Let

$$\mathrm{Br} = \prod_{\xi < \omega_1}^{\mathrm{fs}} S_\xi$$

be the finite-support branch product, computed in L, and let $H \subseteq \mathrm{Br}$ be L-generic. For every $\xi < \omega_1$ set

$$b_\xi = \bigcup \{p(\xi) : p \in H \text{ and } \xi \in \mathrm{dom}(p)\}.$$

Then b_ξ is a cofinal L-generic branch through S_ξ. In this paper, to *destroy* a Suslin tree always means to add such a cofinal generic branch through it. We define

$$W = L[H].$$

Theorem 2.7. *The model W has the following properties.*

1. Br *is a c.c.c. forcing of size* $\aleph_1$*. Hence W preserves all cardinals and cofinalities of L, and $W \models \mathsf{CH}$.*
2. *Every S_ξ is destroyed in W, witnessed by the branch b_ξ.*
3. *For $A \subseteq \omega_1$ in L, let $H_A = H \cap \mathrm{Br}_A$. Then H_A is Br_A-generic over L, and*
$$L[H_A] \models \text{“}S_\xi \text{ has a cofinal branch”} \quad \iff \quad \xi \in A.$$
4. *Every real in W belongs to $L[H_A]$ for some countable $A \subseteq \omega_1$ belonging to L.*

Proof. The first assertion follows from Lemma 2.2. Since $L \models \mathsf{CH}$ and Br has size $\aleph_1$, there are only $\aleph_1$ many nice Br-names for reals, and therefore W still satisfies CH. The definition of H and the dense sets requiring a condition to reach a prescribed level of a prescribed coordinate give (2).

For (3), use the factorization

$$\mathrm{Br} \cong \mathrm{Br}_A \times \mathrm{Br}_{\omega_1 \setminus A}.$$

If $\xi \in A$, then $b_\xi \in L[H_A]$. If $\xi \notin A$, then Lemma 2.2 shows that Br_A preserves S_ξ as a Suslin tree, so no cofinal branch through S_ξ belongs to $L[H_A]$.

Finally, let $\dot{x}$ be a nice Br-name for a real. The antichains occurring in $\dot{x}$ are countable, and the union of the finite supports of all conditions appearing in them is therefore a countable set $A \subseteq \omega_1$ in L. Thus $\dot{x}$ is already a Br_A-name, proving (4). □

The relevant inner models of W used below are obtained by retaining selected branch coordinates from H.

Coding forcing. Put $\Omega = \omega_1^L$. Let $u \in 2^\omega$ and let g be the generic subset of Ω at a fresh $\mathbb{C}_L(\omega_1)$ coordinate. Let $\varrho_L : ({}^{<\Omega}2)^L \to \Omega$ be the canonical bijection. Put

$$h(g) = \{\varrho_L(g \restriction \gamma) : \gamma < \Omega\}, \qquad I(u,g) = \{\omega\gamma + 2k + u(k) : \gamma \in h(g),\ k < \omega\}.$$

Let $X \subseteq \Omega$ canonically code $(u, h(g), \langle b_\xi : \xi \in I(u,g)\rangle)$. We describe the localization $Y = Y(X)$ used by $\mathrm{Code}(u,\eta)$, where η specifies the coordinate supplying g.

Use a recursive theory T consisting of ZFC^- with Collection and Choice, existence of ω_1, the relativizations of these axioms to $L[a]$ for each real a, and $\mathrm{HC}^L = L_{\omega_1^L}$. For a transitive $N \models T$ with $u \in N$ and $\omega_1^N = (\omega_1^L)^N = \nu$, put

$$\mathrm{Pat}(N,u,\nu) \quad\Longleftrightarrow\quad N \models \forall \beta < \nu\ \exists \gamma \in [\beta,\nu)\ \ \mathrm{Pattern}(\gamma, u).$$

Here $\mathrm{Pattern}(\gamma,u)$ asserts that for every $k < \omega$ there is a cofinal branch through $S_{\omega\gamma+2k+u(k)}$. The predicate includes the displayed requirements on N, u, ν. The trees are defined by Lemma 2.6.

In $L[X]$, collapse an elementary submodel of a sufficiently large H_χ of size Ω containing X, u, Ω and all ordinals below Ω. Its collapse N satisfies $\mathrm{Pat}(N,u,\Omega)$. Choose a bijection $e : \Omega \to N$ and code membership and satisfaction via e by $Z \subseteq \Omega$, marking the indices of u and Ω. Use finite tuple codes ordered by maximum. In $L[Z]$, take the club of closure points ν at which the code restricts correctly and $e[\nu] \prec N$, with $u, \Omega \in e[\nu]$ and $e[\nu] \cap \Omega = \nu$. Such a club is obtained by closing under the Skolem maps, e and its inverse. At these points the collapse N_ν satisfies $\mathrm{Pat}(N_\nu, u, \nu)$. Let E consist of the nonzero limit points of this club and 0. If $c \in E$, write $d_c = \min(E \setminus (c+1))$ and let w_c be the $<_L$-least real coding a well-order of type d_c. Set

$$Y(X) = \{2\cdot\xi + 1 : \xi \in Z\} \cup \{2\cdot(c+k) : c \in E,\ k < \omega,\ w_c(k) = 1\}.$$

The intervals $[c, c+\omega)$ are disjoint. Make the choices in $L[X]$ and $L[Z]$ by their canonical well-orders, so the same X gives the same Y in models with the correct Ω. This is the localization argument of David [3]. The second factor of $\mathrm{Code}(u,\eta)$ is $\mathbb{A}_D(Y(X))$.

Let $\Psi(r,u)$ assert that for every countable transitive $M \models T$ with $r, u \in M$ and $\omega_1^M = (\omega_1^L)^M$, one has

$$M \models \exists N \in L[r]\ \ \mathrm{Pat}(N, u, \omega_1^M).$$

A real coding M can also code its satisfaction relation. The well-foundedness requirement is Π^1_1 and the remaining tests on the code are arithmetic. Hence Ψ is Π^1_2, and $\Phi(u) \iff \exists r\, \Psi(r,u)$ is Σ^1_3.

Lemma 2.8. *The generic real r of the almost disjoint factor of* $\mathrm{Code}(u,\eta)$ *satisfies* $\Psi(r,u)$*. Conversely, in an extension preserving* Ω*, if* $\Psi(r,u)$ *holds, then for unboundedly many* $\gamma<\Omega$ *and every* $k<\omega$*,* $L[r]$ *contains a cofinal branch through* $S_{\omega\gamma+2k+u(k)}$*.*

Proof. Given a countable transitive M as in the definition of Ψ, put $\nu=\omega_1^M$. The definition of D and $\mathrm{HC}^L=L_{\omega_1^L}$ give the correct $D\restriction\nu$ in M. The almost disjoint real decodes $Y\cap\nu$ in $L[r]^M$. If $\nu\notin\mathrm{Lim}(E)$, let $c=\max(E\cap\nu)$. Then $c+\omega<\nu\le d_c$. The even part gives $w_c\in M$, contrary to $\nu=\omega_1^M$. At a limit point of E the odd part yields the code of N_ν, so the collapse lies in $L[r]^M$ and is as required.

For the converse, use $F=H_\chi^{L[r,u]}$ for sufficiently large regular χ. This model satisfies T and computes ω_1 and ω_1^L as Ω. If the conclusion in the definition of Ψ failed at F, a countable elementary submodel of F containing r,u,Ω would collapse to a countable transitive model witnessing failure. Thus F sees such an $N\in L[r]$. The trees at Ω are computed correctly, and the branches in N belong to $L[r]$. Finally, once r witnesses $\Phi(u)$ it continues to witness it by Shoenfield absoluteness. □

2.4 Coding iterations and their initial segments

For $E\in L$ of size at most Ω, put

$$C_E=\mathrm{Fn}_{<\Omega}(E\times\Omega,2)^L.$$

Use the Cohen index sets $E_i=[\Omega\cdot i,\Omega\cdot(i+1))$ and put $E_{<\lambda}=\bigcup_{i<\lambda}E_i$. At stage i we first force with C_{E_i} and then with an almost disjoint factor $\dot A_i$, which may be trivial. Thus P_i is the initial segment through the first i stages, and $P_i*C_{E_i}$ is the forcing just before $\dot A_i$.

Whether to code at stage i, the real u_i to be coded, the index $\eta_i\in E_i$, and the isomorphism $\varphi_i:C_{E_i}\cong C_{E_i}\restriction c_i$ are given by P_i-names. Each possible value of φ_i is in L and is induced by bijections on the individual Cohen coordinates, as in the proof below, with at most countably many nonidentity coordinates. If K_i is C_{E_i}-generic, let g_i be the η_i-th Cohen subset of Ω obtained from the filter generated by $\varphi_i[K_i]$. At a coding stage put

$$I_i=I(u_i,g_i),\qquad X_i=\mathrm{code}\big(u_i,h(g_i),\langle b_\xi:\xi\in I_i\rangle\big)$$

and use $\mathbb{A}_D(Y(X_i))$, possibly restricted below a condition a_i. The name for a_i is a $P_i*C_{E_i}$-name. At other stages we still use C_{E_i}, but $\dot A_i$ is trivial and $I_i=\varnothing$.

For $\lambda\le\Omega$, write the forcing as

$$P_\lambda=C_{E_{<\lambda}}*\dot R_\lambda,\qquad R_\lambda=\operatorname*{fs}_{i<\lambda}{}^{*}\,\dot A_i. \tag{1}$$

Here the names for $\dot{A}_i$ are translated from $P_i * C_{E_i}$. The later Cohen forcings are ground-model partial orders and the names for the earlier factors use only the earlier Cohen coordinates. Moreover the order on $\mathbb{A}_D(Y) \restriction a$ is determined by Y and a. Thus the later Cohen forcings can be moved before the earlier almost disjoint factors. At a limit, (1) uses the countable-support Cohen product computed in L and the finite-support almost disjoint iteration. For $\lambda = \Omega$, the Cohen product is isomorphic to the full reservoir.

Use the dense presentation of R_λ in which a condition is a finite function f assigning pairs in $[\omega]^{<\omega} \times [\Omega]^{<\omega}$ to stages in its support. An omitted stage denotes the greatest condition of its factor. At a successor one strengthens the preceding condition to decide the name for the next pair. At a limit a finite support is bounded. This proves density by induction. The same induction shows that the presentations agree in two nested models with the same ordinals if the factors have the same conditions and order in common generic extensions by the preceding factors. At a successor, opposite forcing decisions about membership or order of a pair would contradict agreement in a generic extension of the larger model. At a limit each such decision is made in a bounded initial segment.

We retain the complete embeddings of initial segments and the corresponding translations of names as part of the presentation. A tail in this presentation has a fixed countable length and uses Cohen coordinates disjoint from those of the preceding iteration. The bijections of Cohen index sets used to present products and quotients belong to L.

Lemma 2.9. *The following operations preserve the form* (1)*.*

1. *For $\beta \leq \lambda$, the embedding $P_\beta \lessdot P_\lambda$ is complete and its quotient over the P_β-extension is $C_{E_{<\lambda} \setminus E_{<\beta}} * \dot{R}_{[\beta,\lambda)}$. For a condition (c, f) in the dense presentation the projection to P_β is $(c \restriction (E_{<\beta} \times \Omega), f \restriction \beta)$.*

2. *A tail defined below a nonzero Boolean value d can be lifted to the whole preceding forcing by taking trivial almost disjoint factors off d. Below a condition in the dense presentation a tail again has the form* (1)*, with each initial segment restricted below the projection of the condition.*

3. *Let Q, R be tails over the same preceding forcing. Their product has presentations in both orders on disjoint Cohen index sets. With R first, the initial segment obtained by adjoining Q_i is $R \times Q_i$. Suppose this product is already presented and a Q_i-name for a continuation of fixed countable length is given below a Boolean condition. Lift it by (2) and denote the resulting name by $\dot{T}$. Translating $\dot{T}$ to a name $\dot{T}_R$ over $R \times Q_i$ gives*

$$(R \times Q_i) * \dot{T}_R \simeq R \times (Q_i * \dot{T})$$

over $R \times Q_i$. The previously specified complete embeddings and their translations of names are unchanged.

Proof. For (1), split the Cohen index set into the coordinates used before β and the remaining coordinates. This gives the stated quotient. Any strengthening of the stated projection is compatible with (c, f), by adjoining the later Cohen restrictions and the later values of f. These values remain legal because the conditions at the earlier stages have only been strengthened. This applies at limit stages as well.

For (2), use the given names below d and the trivial almost disjoint factors off d. To restrict below (c, f), write c_η for the partial function at Cohen coordinate η. For each η in the support of c, choose in L a countable ordinal $\nu > \sup(\operatorname{dom} c_\eta) + \omega$ and a bijection from ν onto $\nu \setminus \operatorname{dom} c_\eta$, extended by the identity above ν. These bijections, followed by adjoining c, give an isomorphism $C_E \cong C_E \restriction c$. Compose the isomorphisms on the blocks E_i with the original φ_i, and restrict each almost disjoint factor in the support of f below $f(i)$. The resulting g_i, X_i, and $Y(X_i)$ are the original ones below (c, f). The isomorphisms agree at every initial segment by (1). At each Cohen coordinate they act only on a bounded subset of Ω, and they are closed under composition.

For (3), choose disjoint Cohen index sets for the two tails and translate the names for each tail from its own initial segments. In either order the conditions of the almost disjoint factors and their orders are the same. The preceding induction on the dense presentations applies also at limits. To extend a specified presentation of $R \times Q_i$, retain its two index maps and extend the map for Q_i to the Cohen coordinates of the continuation. The Boolean value used in (2) is translated by the given embedding of Q_i. Thus all embeddings already specified, and their action on names, are preserved. The same calculation translates names along an isomorphism of presentations and its inverse. □

Lemma 2.10. *For $\lambda \leq \Omega$, P_λ has density at most Ω and preserves cardinals and* CH*. In the length-Ω extension every real belongs to the extension by some P_β, $\beta < \Omega$.*

Proof. The Cohen product has size at most Ω and is countably closed in L. Since the branch product is c.c.c., Easton's lemma gives that the Cohen product adds no countable sequences of ordinals over W. In the extension by the Cohen product the finite-support almost disjoint iteration is c.c.c. and has a dense set of size at most Ω. This gives the density, cardinal preservation and CH assertions.

For localization, take a countable nice name for the real in the dense presentation of R_Ω. Its countable ordinal code belongs to W by Easton's lemma, and its support is contained in some $\beta < \Omega$. The dense presentation of R_β computed from $C_{E_{<\beta}}$ agrees with the one computed from the full Cohen product. Thus the same name gives the real in the P_β-extension. □

Lemma 2.11. *In the extension by any iteration of the form* (1)*,*

$$\Phi(u) \quad \iff \quad \exists i < \lambda \ (i \text{ is a coding stage and } u_i = u).$$

Proof. Lemma 2.8 gives the implication from right to left, including when the almost disjoint factor is restricted below a condition. For the converse, let r witness $\Phi(u)$ in the final extension. First work in $W[K]$, where K is generic for $C_{E_{<\lambda}}$. Use countable antichains in the dense presentation to give names for r, u and for the data at each stage, including a_i. Close the union of the supports of the names for r, u under the supports of the names at stages already included. This gives a countable $J \subseteq \lambda$. The names for r, u and for the parameters at stages in J have a common countable ordinal code. By Easton's lemma the code belongs to W, and therefore has a branch-product name of countable support $B \subseteq \Omega$.

The subiteration on J embeds completely into the full iteration. At each stage in J its names depend only on the preceding stages in J. Over the extension by this smaller initial segment the next factor is independent of the quotient of the larger initial segment. This extends the complete embedding at successors. At limits the projections agree on earlier algebras and each finite condition has bounded support. The induced generic for the subiteration gives the same values to the retained names.

Suppose $\xi \notin B \cup \bigcup_{i \in J} I_i$ for the given generic. The model $L[K][H \restriction (\Omega \setminus \{\xi\})]$ contains the ordinal code for the names, where H is the branch generic. In this model form the subiteration on J, except that at the first stage requiring the omitted branch use a trivial factor and continue with trivial factors thereafter. Use a trivial factor also if the corresponding name does not give a unique choice. The test for requiring the omitted branch uses the names, the Cohen generics and I_i, before X_i is formed. The dense presentation of this modified iteration is therefore the same in the smaller model and in $W[K]$. In $W[K]$ it embeds completely into the subiteration on J. The two agree before the first stage requiring the omitted branch, and thereafter the modified iteration has trivial factors. Their restrictions to the Boolean values that no stage requires the omitted branch are isomorphic by induction, taking meets at limits. The given generic contains this Boolean value. It therefore induces a generic for the modified iteration over the smaller model, with the same r, u. The tree S_ξ is Suslin in that extension. Indeed, the Cohen product preserves the c.c.c. of the ground-model branch products, the product of the other branch forcings with S_ξ is c.c.c., and all nontrivial factors of the modified finite-support iteration are σ-centered.

For distinct coding stages i, j, the sets $h(g_i), h(g_j)$ have bounded intersection. The underlying Cohen subsets at distinct coordinates differ on an unbounded set, by density, and changing them on bounded sets preserves this property. Their common initial segments have bounded length, so their images under ϱ_L form a countable set. Suppose no stage codes u. By

Lemma 2.8, choose γ above B and the bounds on all such intersections for pairs in J such that $L[r]$ has branches through every $S_{\omega\gamma+2k+u(k)}$. At most one coding stage in J has $\gamma \in h(g_i)$. If there is one, choose k with $u_i(k) \neq u(k)$. Otherwise take any k. Then

$$\xi = \omega\gamma + 2k + u(k) \notin B \cup \bigcup_{i \in J} I_i.$$

The extension constructed with this branch omitted contains r and hence $L[r]$. This contradicts that S_ξ is Suslin there. □

Up to the specified presentations, the bounded iterations, their restrictions below conditions, and the names for their tails form a set. Each dense set can be indexed by a subset of Ω, each Boolean value is represented by a subset of that dense set, and the names for the parameters at the stages and for the complete embeddings can be given by Boolean-valued functions. For a tail whose countable length is given by a name, first work below a condition deciding the length and apply Lemma 2.9(2). Thus the Boolean values used below may be indexed by tail names of fixed countable length.

2.5 Motivation for the allowability hierarchy

Given the notion of allowable forcing, we now form successive approximations to the class of forcings which we eventually want to use. We first describe the idea leading to these approximations.

Fix a recursive enumeration

$$\langle \varphi_m(v_0, v_1) \mid m \in \omega \rangle$$

of the Π^1_3-formulas with two free real variables, and put

$$A_m := \{(x, y) \in (2^\omega)^2 : \varphi_m(x, y)\}.$$

We write f_m for the uniformizing function which we intend to define for A_m, and write $f(m, x)$ for $f_m(x)$. The goal is to choose, for every $m \in \omega$ and every real x for which

$$A_{m,x} := \{y \in 2^\omega : (x, y) \in A_m\}$$

is nonempty, a real $f(m, x)$ such that

$$(x, f(m, x)) \in A_m,$$

and to code every triple (x, y, m) such that

$$y \neq f(m, x) \quad \text{and} \quad (x, y) \in A_m$$

into $\vec{S}$. The final forcing will have length ω_1, and each of its countable initial segments will be allowable. Since the statement

$$\text{“}(x, y, m) \text{ is coded into } \vec{S}\text{”}$$

is Σ^1_3, the requirement

$$(x, y) \in A_m \quad \text{and} \quad (x, y, m) \text{ is not coded into } \vec{S}$$

is Π^1_3. The bookkeeping of the final iteration will ensure that, whenever $A_{m,x} \neq \varnothing$, exactly one such y remains.

The idea behind the definition can be seen in the following toy example. Work in W, let $x \in W$, and suppose that

$$A_{m,x} = \{y_0, y_1\}.$$

Assume in addition that no allowable forcing over W adds a new member to $A_{m,x}$. We consider only the problem of defining $f(m, x)$ and omit, for the moment, the interaction with the other sections.

Call $y \in A_{m,x}$ *eliminable* if there is an allowable forcing $\mathbb{P}$ such that

$$\mathbb{P} \Vdash (x, y) \notin A_m.$$

Call y *persistent* if it is not eliminable. Thus, if y_0 is persistent, we may reserve y_0 as the value of $f(m, x)$ and code (x, y_1, m) into $\vec{S}$. If y_0 is eliminable, witnessed by an allowable forcing $\mathbb{P}$, we consider y_1. If y_1 is persistent, we reserve y_1 and code (x, y_0, m).

It remains to consider the case in which both values are eliminable. Let $\mathbb{P}$ and $\mathbb{Q}$ witness, respectively, that

$$\mathbb{P} \Vdash (x, y_0) \notin A_m \qquad \text{and} \qquad \mathbb{Q} \Vdash (x, y_1) \notin A_m.$$

By Lemma 2.9, the product

$$\mathbb{P} \times \mathbb{Q}$$

is isomorphic to an allowable forcing. Using this presentation of the product, upward absoluteness gives

$$\mathbb{P} \times \mathbb{Q} \Vdash (x, y_0) \notin A_m \wedge (x, y_1) \notin A_m.$$

By the assumption that no new member is added to the section, the product forces

$$A_{m,x} = \varnothing.$$

In this case there is no value to uniformize at x.

This argument isolates the two operations which are needed later:

reserve a value which is persistent under the forcings under consideration;
combine eliminating forcings by taking products. (2)

The difficulty is that an eliminating forcing may add codes which interfere with other sections. Thus it is not enough to use arbitrary allowable forcings as witnesses. We need a class $\mathcal{P}$ of allowable forcings such that the following rule produces forcings in $\mathcal{P}$ again. For every task (x, m) considered by the bookkeeping, either

$$\exists y \in A_{m,x}\ \forall \mathbb{Q} \in \mathcal{P}\big(\mathbb{Q} \text{ extending the current forcing} \to \mathbb{Q} \Vdash (x, y) \in A_m\big),$$

in which case one such y is reserved and the other values are coded, or every value under consideration can be eliminated by a forcing in $\mathcal{P}$.

This turns the construction into a fixed point problem. Let

$$\mathcal{A}_0^W$$

be the class of allowable iterations. The first refinement searches for values which are persistent under $\mathcal{A}_0^W$-extensions; the resulting iterations form $\mathcal{A}_1^W$. The next refinement searches also for values which are persistent under $\mathcal{A}_1^W$-extensions, and so on. We obtain a decreasing hierarchy

$$\mathcal{A}_0^W \supseteq \mathcal{A}_1^W \supseteq \cdots \supseteq \mathcal{A}_\alpha^W \supseteq \cdots .$$

In the first refinement we keep a minimizer uncoded whenever all values that can attain the least rank are already present.

The reason for continuing the construction is the usual moving-target problem. In the definition of 1-allowability we test persistence under 0-allowable extensions, but the iteration which we produce is itself 1-allowable. We should therefore also test persistence under 1-allowable extensions. Repeating this argument leads to the transfinite hierarchy defined in the next subsection.

2.6 Persistence and the choice of a uniformizing value

Use the coding iterations of Subsection 2.4, with the specified complete embeddings of their initial segments. We work in extensions M by such an initial segment, sometimes restricted below a condition. An iteration over M is a quotient with the presentation given in that subsection. The phrase *code* (m, x, y) means to code the real $\langle x, y, m\rangle$. Put

$$A(m, x, y) \quad \Longleftrightarrow \quad (x, y) \in A_m.$$

We use two absoluteness facts for A. A witness to $\neg A(b)$ in the extension by one factor of a product remains a witness in the product extension. If $\neg A(b)$ holds in the final extension, its witness gives $\neg A(b)$ in each sufficiently late bounded initial segment of the iteration. Here $b = (m, x, y)$ belongs to the model in which membership is tested. In the construction over L, write $\neg A(b)$ as $\exists z\, \psi(b, z)$ with $\psi \in \Pi^1_2$ and apply Shoenfield absoluteness to ψ. The second assertion also uses the fact that every real belongs to an extension by a bounded initial segment.

Choice functions. For each extension M under consideration we include a name for a choice function J_M on its nonempty sets of reals, so $J_M(B) \in B$. Along each specified inclusion $M \subseteq N$, require $J_N(B) = J_M(B)$ for $B \in \mathrm{dom}(J_M)$. These are names over the given forcings.

Lemma 2.12 (Choice functions in product extensions)**.** *Let Q, R be countable iterations over the same M, with choice functions on the extensions by their initial segments agreeing at M. There are choice functions on the extensions by all products $Q_i \times R_j$ of initial segments. They extend the given choice functions and agree along the specified inclusions. One can retain a prescribed such family for the products of initial segments of Q_{i_0} and R_{j_0}. The same assertions hold when passing to restrictions below conditions, when defining an iteration by cases below a Boolean value and its complement, and when passing to quotients by products of initial segments. They also hold when a further iteration is appended to one factor of a specified product.*

Proof. In a product extension write $M_{ij} = M[G_{Q_i}, G_{R_j}]$. For incomparable pairs (i, j) and (i', j'), factor over $M_{\min(i,i'),\min(j,j')}$. Generic intersection for the two independent quotients implies that any set of reals B in both extensions belongs to this smaller extension. Indeed, each element of B belongs to the smaller extension by generic intersection. A well-order of its reals then identifies B with a subset of an ordinal, to which generic intersection applies again. It follows that each B has a least pair (i, j) at which it appears. Choose a pair witnessing the minimum in each coordinate and take the coordinatewise minimum of those pairs.

At the least pair for B, use the prescribed choice if there is one. Otherwise choose from B by a well-order named over $Q_i \times R_j$. Use this choice at every larger pair. The least pair of a set in the prescribed family stays the same when the factors are extended. Taking its coordinatewise minimum with any new pair at which the set occurs gives a pair in the prescribed family, by generic intersection. When considering only pairs $(i, j) \geq (i_0, j_0)$, the least pair for an old B is the coordinatewise maximum of its original least pair and (i_0, j_0). If one coordinate is i_0 or j_0, the choice is already given on the corresponding factor. This proves compatibility when the product is factored in either order and when passing to a quotient by $Q_{i_0} \times R_{j_0}$. These choices can be made by names, since membership in a smaller extension is expressed by a disjunction over names lifted from its forcing.

Below a condition use the names induced by the specified complete embeddings. When defining an iteration by cases below a Boolean value d and its complement, use the prescribed choice functions below d and extend them coherently on the complement. A condition in the dense presentation of the product is a pair of conditions in the two factors. Restricting below it therefore gives the choice functions on the corresponding restrictions of the two factors and their initial segments. If a further iteration over $M[G_{Q_{i_0}}]$ is defined below d, extend its name to the full forcing by using trivial almost

disjoint factors off d. Extend the choice functions to products with the longer first factor by the construction above. This retains the earlier choices and complete embeddings. □

We work with the set of all countable iterations under consideration, together with their specified embeddings and the names for their choice functions. The same forcing may occur with different data. For a presentation of density at most Ω, the regular open algebra has size at most 2^{Ω}, and names for reals are coded by countable sequences of its elements. A name for a set of reals gives the Boolean value of membership for each such real name. The same description, applied to the graph, gives names for the choice functions. Thus the presentations together with the names and embeddings still form a set, possibly of size greater than 2^{Ω}. This set includes the restrictions and products in Lemma 2.12. A name for a quotient iteration defined below a condition can be used there or extended by trivial almost disjoint factors on the complement, with the same countable length.

The allowability hierarchy. We define successive hierarchies of allowable iterations. Each hierarchy begins at a specified initial segment of the final iteration. The ξ-th hierarchy retains the restrictions on coding imposed so far and the fixed classes from hierarchies $\eta < \xi$. A restriction forbids coding a specified triple (m, x, y). At a later extension by a bounded initial segment, read the earlier definitions using the complete embeddings from the initial segments at which those definitions began. In a product with a further iteration these embeddings are obtained through the factor belonging to the final iteration. Only the construction of the final iteration begins a new hierarchy.

For an extension M considered in the current hierarchy, let $\mathcal{K}_0^{\xi}(M)$ consist of the quotient iterations over M that obey all the restrictions on coding and belong to every earlier fixed class, evaluated over M. Trivial almost disjoint factors are permitted. Given $\mathcal{K}_{\beta}^{\xi}$ in all these extensions, put

$$P_{\beta}^{\xi}(M) = \{b \in (\omega \times \mathbb{R}^2)^M : \forall Q \in \mathcal{K}_{\beta}^{\xi}(M) \quad Q \Vdash_M A(b)\}.$$

The rank through (ξ, β) of b is the first pair, in lexicographic order, at which b belongs to a persistence set. Use all levels of the earlier hierarchies up to their fixed levels and the levels at most β of the current hierarchy. Write $\rho_{\leq(\xi,\beta)}(b)$ for this partial rank. It is undefined precisely when b has no rank at any of these levels, including those in the earlier hierarchies.

For $m, x \in M$, let $\Lambda_{\beta}^{\xi}(M, m, x)$ be the set of pairs λ for which some $Q \in \mathcal{K}_{\beta}^{\xi}(M)$, some $q \in Q$, and some real name $\dot{y}$ satisfy

$$q \Vdash_Q^M \rho_{\leq(\xi,\beta)}(m, x, \dot{y}) = \lambda.$$

When this set is nonempty, put $\lambda = \min \Lambda^{\xi}_{\beta}(M, m, x)$. Call this minimum *stable* if no condition witnessing it forces $\dot{y} \notin \mathbb{R}^M$. In this case let B be the nonempty set of $y \in \mathbb{R}^M$ for which some such condition forces $\rho_{\leq(\xi,\beta)}(m, x, \check{y}) = \lambda$, and put $j^{\xi}_{\beta}(M, m, x) = J_M(B)$. Thus j^{ξ}_{β} is defined precisely in the stable case.

An iteration over M belongs to $\mathcal{K}^{\xi}_{\beta+1}(M)$ if it belongs to $\mathcal{K}^{\xi}_{\beta}(M)$ and, whenever a stage codes (m, x, y), the extension N by the preceding initial segment satisfies

$$j^{\xi}_{\beta}(N, m, x) \downarrow \implies y \neq j^{\xi}_{\beta}(N, m, x). \tag{3}$$

Take intersections at limit levels. At a limit of hierarchies retain the union of the restrictions on coding and use every earlier fixed class in the definition at level zero.

Lemma 2.13 (Boolean values and stabilization)**.** *At the beginning of each hierarchy, the recursion defines Boolean values for all the named iterations and their quotients under consideration. These values agree under the translations induced by the specified embeddings, including in quotients evaluated over product extensions. The hierarchy stabilizes at some ordinal θ_{ξ}.*

Proof. Let H be a named iteration over the model in which the hierarchy begins, and let B_H be its Boolean algebra. Write $a_{\beta}(H, \dot{T})$ for the Boolean value in B_H that $\dot{T}$ belongs to the class at level β. For a complete embedding $j : B_H \longrightarrow B_{H*\dot{T}}$, write $\exists_j c$ for the least $d \in B_H$ with $c \leq j(d)$ and put $\forall_j c = \neg\exists_j \neg c$. These give the Boolean values that the image of c in the quotient is nonzero or is the top element, respectively. Use the relative Boolean operations below a condition.

At level zero take the Boolean value of the conjunction of the restrictions on coding and membership in all earlier fixed classes. Given the Boolean values at level β, the Boolean value of $\dot{b} \in P^{\xi}_{\beta}$ is

$$\bigwedge_{\dot{T}} \left(\neg a_{\beta}(H, \dot{T}) \vee \forall_{j_{\dot{T}}} \| A(\dot{b}) \|_{H*\dot{T}} \right).$$

After forming these values for every H, the Boolean value that λ is attained in an allowable extension is

$$\bigvee_{\dot{T}, \dot{y}} \left(a_{\beta}(H, \dot{T}) \wedge \exists_{j_{\dot{T}}} \| \rho_{\leq(\xi,\beta)}(m, \dot{x}, \dot{y}) = \lambda \|_{H*\dot{T}} \right).$$

Adding $\dot{y} \notin \dot{\mathbb{R}}^H$ inside the forcing statement gives the Boolean value for attaining λ with a new real. Requiring $\dot{y}$ to equal a lifted name from H gives the Boolean values of membership in B. Leastness and the name for J_H then give the Boolean value of (3). If g_i is this value in the algebra of the

initial segment preceding stage i, with $g_i = 1$ when the almost disjoint factor is trivial, put

$$a_{\beta+1}(H, \dot{T}) = a_\beta(H, \dot{T}) \wedge \bigwedge_i \forall_{j_i} g_i.$$

This enforces the requirement on every branch. At a limit take the meet of the preceding families of Boolean values.

For an earlier hierarchy, if $H = A * P$ factors through the initial segment at which a later hierarchy begins, the value over P is the quotient image of the value over H by G_A. Indeed, take names for the iteration and its quotient below a condition in G_A. Form the two-step iteration over the model in which the earlier hierarchy began, using the specified complete embeddings to translate the names. Then

$$a_\beta(H, \dot{T}) \in G_A * G_P \quad \Longleftrightarrow \quad a_\beta(H, \dot{T})/G_A \in G_P.$$

Names equivalent below a condition give the same value there. Likewise the translations preserving the earlier restrictions, embeddings, and choice functions preserve the values. This follows by induction from the complete embeddings, their projections, and the Boolean operations above. If two names agree in the generic extension, first strengthen the condition to force the agreement. In a product use the earlier embeddings through the factor belonging to the final iteration. Appending a further iteration leaves these embeddings fixed. At a limit of hierarchies the earlier families of values form a set, and one takes the meet of the corresponding quotient values.

The families of Boolean values decrease in a fixed set, so two consecutive families eventually agree. At that level the persistence sets, ranks, minima and choices do not change at the next level. Thus the same successor condition applies and the families remain constant. Denote the fixed class by $K_\xi(M) = \mathcal{K}^\xi_{\theta_\xi}(M)$. □

Lemma 2.14 (Products and preservation of ranks)**.** *If $Q, R \in \mathcal{K}^\xi_\alpha(M)$, then the quotient of $Q \times R$ over the extension by either factor is in the class at level α there. For an extension by an iteration in this class, the partial rank through (ξ, α) of a triple in M is preserved, including whether it is defined. For each (m, x) in M, the set of attainable ranks at level α is preserved, as is the stability of its minimum. If the minimum is stable, the set of minimizers and its choice are also preserved.*

Proof. Induct on ξ and then on α, proving the assertion about products first. Restrictions below conditions, initial segments and quotients by initial segments preserve the requirements. An iteration defined by cases below d and $\neg d$ satisfies the given requirements below d and has trivial almost disjoint factors elsewhere. For a concatenation of allowable iterations check each coding stage in the extension by its preceding initial segment. These facts follow from the definition and Lemma 2.13.

In the product, the real coded at a stage of one factor is given by the name from that factor. Thus the restrictions on coding still hold. The assertions about products for the earlier hierarchies give membership in their fixed classes, with the specified embeddings from the initial segments at which they began. To check (3) at a level $\beta < \alpha$, compare the extension by the preceding initial segment of the factor with its further extension by the other factor. This further forcing is allowable at level β by induction, so the stability of the minimum and the choice in (3) are preserved. This proves the assertion about products at level α.

Let $N = M[G_R]$, where $R \in \mathcal{K}^{\xi}_{\alpha}(M)$. If an allowable iteration over N has a condition forcing $\neg A(b)$ for $b \in M$, decide its name and length below a condition in G_R and use trivial almost disjoint factors on the complement. This gives a concatenation over M with a condition forcing $\neg A(b)$. Conversely, if an allowable iteration Q over M has such a condition, take the product $Q \times R$. Its quotient over N is allowable at level α and the witness to $\neg A(b)$ persists in the product. This proves equality of the persistence sets on the triples of M at level α. The lower levels and the earlier hierarchies give equality of the partial ranks.

A witness that a rank is attained over N can likewise be composed with R. Conversely, take T witnessing that a rank is attained over M, and form $T \times R$. The rank of its real in the extension by T is preserved in the product because the quotient over that extension is allowable at level α. If this real is new over M, it is still new over N by generic intersection. If every real witnessing the least rank attainable over M already belongs to M, composition shows that every minimizer over N belongs to that set, and the product argument gives the reverse inclusion. For a stable minimum λ, the minimizers are exactly the reals of rank λ already in M, by preservation of ranks and the trivial forcing. Thus both extensions use the same choice function entry. $\square$

In particular, whenever (3) from an earlier level or hierarchy forbids coding a triple over M, the minimum remains stable and the same triple is forbidden by the corresponding requirement at the current fixed level. Indeed, a minimizer already in M witnesses λ using the trivial forcing in the smaller class. Any smaller rank attainable in the smaller class would already have been visible at the earlier level, since newly assigned ranks are later in the lexicographic order. Any new real of rank λ obtainable in the smaller class would also have been obtainable in the earlier class. Thus the set of minimizers and its choice agree. If the current minimum can be attained by a new real, no earlier requirement of this kind is defined at (m, x) over M.

Lemma 2.15 (Products after imposing new restrictions on coding)**.** *Let M be an extension by a bounded initial segment of the final iteration during the η-th hierarchy, and let $R \in K_{\eta}(M)$ be a further bounded segment of the final iteration. Suppose that for each triple (m, x, y) forbidden during this segment,*

$y \notin M$. If $Q \in K_\eta(M)$, the quotient of $Q \times R$ over $M[G_R]$ satisfies all the intervening fixed classes and the current fixed class. For each $S = M[G_{Q_i}]$, the partial rank of a triple in S agrees in S at the η-th fixed level and in $M[G_{Q_i}, G_R]$ at the current fixed level. Every minimum attainable over S in the η-th fixed class remains the minimum and keeps its stability. In the stable case the set of minimizers and the choice are preserved.

Proof. Fix the product presentation with the specified embeddings. Induct on the later hierarchies and then on their levels, simultaneously for all $Q \in K_\eta(M)$ and their initial segments. At a given hierarchy, allow any bounded extension by the final iteration containing the initial segment at which that hierarchy began. Evaluate its Boolean values there, including when a later hierarchy is already in use. This lets us evaluate the earlier requirements on the same product. If a further iteration is appended to one factor, the earlier complete embeddings and choice functions are unchanged by Lemmas 2.12 and 2.13. The earlier fixed classes are preserved by Lemma 2.14 and the induction on the hierarchies. No coding stage of Q violates a new restriction on a triple (m, x, y). Otherwise y would belong to the extension by an initial segment of Q and also to $M[G_R]$, contrary to $y \notin M$ and generic intersection. This proves the assertion at level zero. At a limit of hierarchies apply the previous assertions to all the earlier fixed classes.

Assume the product assertion at level β of the hierarchy under consideration. Let $S = M[G_{Q_i}]$ and suppose $b \in S$ has no rank at the η-th fixed level there. Some iteration in $K_\eta(S)$ has a condition forcing $\neg A(b)$. Restrict below that condition. Over a condition in the Q_i-generic, choose its name and length, use trivial almost disjoint factors on the complement, and append it to Q_i. The result is an iteration in $K_\eta(M)$. Take its product with R preserving the specified embeddings for $Q_i \times R$. By the induction at β, the appended iteration in the quotient over $M[G_{Q_i}, G_R]$ is allowable at level β. The witness to $\neg A(b)$ persists in this product. Consequently b does not enter the persistence set at level β. Ranks already assigned at the η-th fixed level agree by Lemma 2.14, with the earlier definitions evaluated in the same product extension.

If there is a minimum attainable over S in $K_\eta(S)$, append a witnessing iteration to Q_i in the same way. A witness new over S remains new over $M[G_{Q_i}, G_R]$ by generic intersection over S. Conversely, R in the product is allowable over S at the η-th fixed level by Lemma 2.14. Composing it with a witnessing iteration in the current class over $M[G_{Q_i}, G_R]$ gives an iteration in $K_\eta(S)$. A smaller rank would already be visible through the η-th hierarchy, contrary to the original minimality. If the old minimum is stable, the same composition shows that all the minimizers are among the old ones. The old minimizers retain their ranks, so the set of minimizers and the choice agree.

These assertions verify (3) for the stages of Q in the quotient at the next

level. When the minimum existed over S, its stability and, when defined, the choice agree. If there was no attainable rank over S, a real to be coded at a stage with preceding extension S is unranked there, since otherwise the trivial forcing would witness an attainable rank for it. The argument with a condition forcing $\neg A(b)$ shows that this real remains unranked at the level in question, so it cannot be chosen by j^{ξ}_{β}. At limit levels use all the earlier requirements. This proves the assertions at the fixed level. □

Lemma 2.16 (First initial segment containing a minimizer)**.** *Suppose the current fixed class allows its minimum rank λ at (m, x) to be attained by a real outside M. There is a restriction of an allowable iteration that forces a real y of rank λ at (m, x) to appear in its final extension and in no extension by a proper initial segment.*

Proof. Take a witnessing iteration Q and a condition forcing that $\dot{y}$ is new and attains the minimum. Strengthen it to decide the least $\delta > 0$ such that $\dot{y}$ equals a lifted Q_δ-name, and decide that name $\dot{z}$. The quotient after δ is allowable in the same hierarchy. By Lemma 2.14, the strengthened condition forces that the extension by Q_δ satisfies

$$\dot{z} \notin \bigcup_{\gamma<\delta} \dot{\mathbb{R}}^{Q_\gamma}, \qquad \rho_{\leq(\xi,\theta_\xi)}(m, x, \dot{z}) = \lambda.$$

Take a condition p in the dense presentation below it. The projection of p to Q_δ forces both statements over Q_δ, since every strengthening of the projection is compatible with p. Restrict Q_δ below this projection, retaining the specified embeddings of the restricted initial segments. This applies also at a limit δ. □

3 The uniformizing extension

Theorem 3.1. *Suppose the coding iterations have the properties in Lemmas 2.9, 2.10 and 2.11, and witnesses to $\neg A(b)$ have the two absoluteness properties in Subsection 2.6. Suppose also that the iterations preserve the regularity of Ω and that $2^{\aleph_0} = \Omega$ in the extensions by their bounded initial segments. Then there is a coding iteration of length Ω in whose extension every A_m is uniformized by*

$$(x, y) \in A_m \ \wedge \ \neg\Phi(\langle x, y, m\rangle).$$

Proof. Let P_i denote the initial segment of length i of the iteration to be constructed. For each $i < \Omega$, fix a P_i-name for an enumeration $\langle x_{i,j} : j < \Omega\rangle$ of the reals in the P_i-extension. The bookkeeping lists each instruction below cofinally often, giving each real by a pair (i, j). Omit an instruction if one of its initial segments has yet to be constructed. Record at most one choice y_* for each (m, x).

1. If $b = (m, x, y)$ has no rank through the current fixed level, append a tail in the current fixed class restricted below a condition forcing $\neg A(b)$.

2. Given (m, x, y), code $\langle x, y, m\rangle$ if the current fixed class permits it.

3. If a value for (m, x) has yet to be chosen and there is a least rank attainable in the current fixed class, proceed as follows. If the minimum is stable, choose $y_* = J_M(B)$, where B is the set of minimizers. Otherwise use Lemma 2.16 to append an allowable tail ending at the first initial segment whose extension contains a new minimizer y_*. Choose y_* and restart the hierarchy there, adding (m, x, y_*) to the list of triples whose codes are forbidden. The new restriction applies before the next stage of the iteration.

Use the Boolean values in Lemma 2.13 to implement these instructions by names. On a maximal antichain in the forcing already constructed, choose a name for the next tail and decide its countable length. At each subsequent stage, mix the names for the corresponding factors of these tails, using the index of that stage within the tail. The length used by a single instruction may vary with the generic filter. At a limit within a tail take the limit of its initial segments. Between instructions extend the union of the earlier choice functions to the nonempty sets of reals that have just appeared.

At a limit of restarts, take the union of the lists of forbidden triples and start with the intersection of the earlier fixed classes. At a restart the new hierarchy is used for the next factor, and the earlier hierarchies still determine the old rank of the chosen value. An instruction that adds no tail is followed by a stage with trivial almost disjoint factor, after making any choice or restarting the hierarchy. Thus each instruction uses at least one stage. In the generic extension, let γ_ν be the stage at which instruction ν begins. Each instruction uses countably many stages and $\gamma_\lambda = \sup_{\nu<\lambda} \gamma_\nu$ at limits. Regularity of Ω gives $\gamma_\nu < \Omega$ for every $\nu < \Omega$, and $\gamma_\nu \geq \nu$. Every instruction involving a finite tuple of reals in the final extension is therefore considered after those reals have appeared.

Verification. Every bounded tail of the iteration is in the fixed class used at its starting stage, since each of its factors satisfies the requirements of all earlier hierarchies. If a later restriction forbids coding $\langle x, y, m\rangle$, then y first appears in the extension at the stage at which that restriction is introduced. By Lemmas 2.14 and 2.15, along the iteration the rank of a triple already present is preserved, as is the absence of a rank.

If a ranked triple failed to belong to A in the final extension, a witness to this failure would already occur in an extension by a bounded initial segment, contrary to persistence. An unranked triple remains unranked until

instruction (1) applies. The resulting witness to $\neg A$ persists. Hence every final member of A has a rank in each intermediate extension that contains it.

Suppose $A_{m,x}$ is nonempty in the final extension. Some intermediate extension contains a ranked member, so the trivial forcing witnesses an attainable rank. The least attainable rank is preserved and instruction (3) eventually chooses a value y_*. Since y_* has a rank, it belongs to the final section.

Suppose first that y_* was chosen at a stable minimum. If $\langle x, y_*, m\rangle$ had been coded earlier, then x, y_* would already have been present at that stage, and y_* would have had a rank there. The trivial forcing would then have witnessed an attainable rank. The least attainable rank, its stability and the choice of a minimizer are preserved up to the stage of selection. Thus the earlier coding would have violated (3). The same restriction applies after the selection. If y_* was chosen as a new minimizer, it was absent from every earlier extension in which a code could have been specified. At the first initial segment whose extension contains y_*, coding $\langle x, y_*, m\rangle$ is forbidden from then on. This argument also applies when that initial segment has limit length.

Let $y \neq y_*$. Consider instruction (2) for (m, x, y) after y_* has been chosen and y has appeared. In the stable case, every earlier requirement based on a stable minimum makes the same choice y_*, by the observation after Lemma 2.14. In the other case apply Lemma 2.15 from the beginning of the instruction that chose y_*, when y_* was still absent. The minimum can still be attained by a new real, so no earlier hierarchy can have a stable minimum at this input. The only triple that can have been explicitly forbidden for this input is (m, x, y_*). Thus instruction (2) codes $\langle x, y, m\rangle$.

By exactness the relation

$$G_m(x, y) \quad \Longleftrightarrow \quad (x, y) \in A_m \ \wedge \ \neg\Phi(\langle x, y, m\rangle)$$

selects exactly one value in each nonempty section. If A_m is Π^1_k and Φ is Σ^1_k, then G_m is Π^1_k. For relations with a real parameter, include the parameter in the first coordinate of a universal relation. □

Corollary 3.2. *There is a cardinal-preserving set-generic extension of L in which* CH *and the Π^1_3-uniformization property hold.*

Proof. First force with the branch product producing W. Apply Theorem 3.1 in W, using the coding predicate of Lemma 2.8 and the iterations of Subsection 2.4. Cardinal preservation and CH follow from Theorem 2.7 and Lemma 2.10. Since Φ is Σ^1_3, the uniformizing graphs are Π^1_3. □

4 Forcing over the canonical inner models M_n

Fix for the remainder of this section a natural number n with $1 \leq n < \omega$, and assume that $M_n^{\#}$ exists. As always we let M_n be the resulting inner model

one obtains when iterating the sharp out of the universe. Put $\Omega = \omega_1^{M_n}$. Our goal is to prove the following theorem.

Theorem. *There is a set-generic extension of M_n in which the Π^1_{n+3}-uniformization property holds.*

The proof follows the construction over L from the preceding sections, but three ingredients require M_n-analogues: the recovery of the canonical lower part $M_n|\omega_1$, the definition of the coding predicate, and the absoluteness argument for witnesses to nonmembership in the sections. We begin with the inner-model-theoretic background needed for these analogues.

4.1 Inner-model-theoretic background and generic absoluteness

We use the notation and comparison conventions of Steel's analysis of mice with finitely many Woodin cardinals; see [20, 21]. An *ordinary premouse* means a premouse in the parameter-free premouse language. In particular, no real or other base set is included as a predicate.

Definition 4.1. *Let $\mathcal{M}$ be a premouse and let $\eta < \mathrm{Ord}^{\mathcal{M}}$. We say that $\mathcal{M}$ is* n-small above η *if, whenever E is an extender on the $\mathcal{M}$-sequence and*

$$\eta < \mathrm{crit}(E),$$

the initial segment $\mathcal{J}^{\mathcal{M}}_{\mathrm{crit}(E)}$ does not have n Woodin cardinals above η. We say that $\mathcal{M}$ is n-small *if it is n-small above* 0.

We use Steel's finite-level iterability predicates with their parity-dependent definitions [20, Definitions 1.4 and 1.6]. For a premouse $\mathcal{M}$ and a cutpoint η, write $\mathrm{It}_n(\mathcal{M}, \eta)$ for this predicate above η. We will use the cutpoint ω.

Steel's analysis gives the following complexity statement.

Fact 4.2. *For each fixed $1 \leq n < \omega$, the relation*

"*x codes a countable premouse which is Π_n-iterable above ω.*"

is Π^1_{n+1} in the real code x. Moreover, every countable realizable n-small premouse is Π_n-iterable at the indicated cutpoint.

We next specify the generic extensions in which the comparison lemma will be used. All cardinalities in the following definition are computed in M_n.

Definition 4.3. *A model N is a* small M_n-generic extension *if*

$$N = M_n[G]$$

for some cardinal-preserving forcing $\mathbb{P} \in M_n$ such that $M_n \models |\mathbb{P}| \leq \aleph_1$ and some M_n-generic filter $G \subseteq \mathbb{P}$. A small two-step extension *is an extension*

$$M_n[G] \subseteq M_n[G][H]$$

obtained from a two-step forcing $\mathbb{P} * \dot{\mathbb{Q}} \in M_n$ which is forced to be cardinal preserving and of size at most $\aleph_1$.

The preparatory branch forcing and every forcing iteration used later will be shown to have size $\aleph_1$ and to preserve cardinals. Hence all models which occur in the proof are small M_n-generic extensions in the sense of Definition 4.3.

The second background ingredient is the mouse-operator closure which yields projective generic absoluteness. Recall that a set is *self-wellordered*, abbreviated *swo*, if it codes a wellorder of itself. For a self-wellordered set X, let $M_k^{\#}(X)$ denote the canonical active X-mouse with k Woodin cardinals. Thus $M_0^{\#}(X)$ is the usual sharp over X.

Fact 4.4. *The model M_n is closed under the operator*

$$X \longmapsto M_{n-1}^{\#}(X)$$

below its least Woodin cardinal. In particular, for every self-wellordered $X \in H(\omega_2)^{M_n}$, the mouse $M_{n-1}^{\#}(X)$ exists and is normally ω_2-iterable in M_n.

This closure is preserved by the small forcing notions which occur in the construction.

Lemma 4.5. *Let $N = M_n[G]$ be a small M_n-generic extension. Then*

$$N \models \text{“}M_{n-1}^{\#}(X) \text{ exists and is normally } \omega_2\text{-iterable for every swo } X \in H(\omega_2)\text{.”}$$

Moreover, whenever X belongs to both M_n and N, the two models compute the same mouse $M_{n-1}^{\#}(X)$.

Proof. Apply the generic preservation theorem for mouse operators to $M_{n-1}^{\#}$. In the notation of [19, Lemma 3.4], take $\kappa = \omega_1$ and replace the index occurring there by $n-1$. Fact 4.4 supplies the hypothesis that $M_{n-1}^{\#}(X)$ exists for every self-wellordered $X \in H(\omega_2)^{M_n}$. The cited lemma gives existence, normal ω_2-iterability, and correctness of the operator in the forcing extension. □

We can now state the precise generic absoluteness principle used later. It is the two-step form which permits real parameters introduced by the first forcing.

Theorem 4.6. *Let*

$$M_n[G] \subseteq M_n[G][H]$$

be a small two-step extension. Then

$$M_n[G] \prec_{\Sigma^1_{n+2}} M_n[G][H].$$

Equivalently, for every real parameter $a \in M_n[G]$ and every Σ^1_{n+2} formula $\vartheta(v)$,

$$M_n[G] \models \vartheta(a) \quad \Longleftrightarrow \quad M_n[G][H] \models \vartheta(a).$$

The same equivalence holds for Π^1_{n+2} formulas.

Proof. By Lemma 4.5, the intermediate model $M_n[G]$ is closed under $X \mapsto M^{\#}_{n-1}(X)$ for self-wellordered $X \in H(\omega_2)^{M_n[G]}$. Apply the Martin–Solovay–Schindler generic absoluteness theorem in the form of [19, Lemma 3.7], with the mouse index $n-1$ and $\kappa = \omega_1$, to the second forcing. This yields Σ^1_{n+2}-elementarity between $M_n[G]$ and $M_n[G][H]$. The Π^1_{n+2} assertion follows by taking complements. □

The point at which Theorem 4.6 enters the uniformization proof is recorded in the following consequence.

Corollary 4.7. *Let*

$$M_n[G] \subseteq M_n[G][H]$$

be a small two-step extension, let $a \in M_n[G]$ be a real, and let $\varphi(v)$ be a Σ^1_{n+3} formula. If

$$M_n[G] \models \varphi(a),$$

then

$$M_n[G][H] \models \varphi(a).$$

Consequently, if $A \subseteq (2^\omega)^2$ is Π^1_{n+3} and

$$M_n[G] \models (x, y) \notin A,$$

then $(x, y) \notin A$ remains true in $M_n[G][H]$.

Proof. Write

$$\varphi(a) \equiv \exists z\, \psi(z, a),$$

where ψ is Π^1_{n+2}. Choose a witness $z \in M_n[G]$. By Theorem 4.6, the statement $\psi(z, a)$ remains true in $M_n[G][H]$, so the same witness establishes $\varphi(a)$ there. The final assertion is the special case in which $\varphi(x, y)$ defines the complement of A. □

4.2 Recognition of the lower part

Definition 4.8. *Let $\mathcal{I}_n$ be the set of real codes for well-founded ordinary premice P such that*

$$P \text{ is } \omega\text{-sound}, \qquad \rho_\omega(P) = \omega,$$
$$P \text{ is } n\text{-small}, \qquad \mathrm{It}_n(P, \omega).$$

Here ω is a cutpoint, and we use Steel's iterability predicate from [20, Definitions 1.4 and 1.6]. We also write $P \in \mathcal{I}_n$ when a real code for P belongs to this set. For an ambient model N, let $\mathcal{C}_n^N$ be the class of transitive premouse structures whose codes in N satisfy the definition in N.

The definition of $\mathcal{I}_n$ is parameter-free Π^1_{n+1} and is invariant under a change of the enumeration of the coded premouse. Indeed the finite iterability predicate is Π_n over HC by [20, Lemmas 1.5 and 1.7], giving the bound in real codes in Fact 4.2. The remaining checks use the satisfaction relations of the coded structures and the finite-degree fine-structural definitions. They do not increase the bound, and well-foundedness is Π^1_1.

Lemma 4.9. *Put $\Omega = \omega_1^{M_n}$. Every member of $\mathcal{C}_n^{M_n}$ is an initial segment of M_n of height below Ω, and their heights are cofinal in Ω. If N is a small M_n-generic extension, then*

$$\mathcal{C}_n^N = \mathcal{C}_n^{M_n}, \qquad \bigcup \mathcal{C}_n^N = M_n|\Omega.$$

Proof. First work in M_n. Its reals are captured by the realizable, sound, projectum-ω segments in Steel's construction [20, Theorem 2.3]; see also [21, Corollaries 3.12–3.13]. These segments are countable in M_n and have unbounded heights in Ω, since a bound would put every real of M_n in a single countable level. In the background for this construction M_n is Σ^{HC}_n-correct by [20, Corollary 4.4]. Thus the iterability predicate, which is Π_n over HC, agrees between M_n and the background on the relevant premice. The realizable segments satisfy the predicate there by [20, Lemma 2.2(2)] and hence belong to $\mathcal{C}_n^{M_n}$.

If $P \in \mathcal{C}_n^{M_n}$, choose a realizable segment Q of strictly greater height from this cofinal system. Both are ω-mice in the sense of [20, Definition 1.3]. They agree below the common cutpoint ω. By [20, Lemma 2.2(3)] they are comparable by initial segment, so $P \trianglelefteq Q \triangleleft M_n$. This proves ground recognition and cofinality.

Now let N be a small extension. For premouse codes p, q, let $\mathrm{Comp}(p, q)$ assert that either structure is isomorphic to an initial segment of the other. This has an analytic presentation: an initial-segment cut and an isomorphism are coded by a real. If $q \in M_n$ codes a recognized segment, then

$$\exists p \, (p \in \mathcal{I}_n \ \wedge \ \neg \, \mathrm{Comp}(p, q))$$

is $\Sigma^1_{n+2}(q)$ and is false in M_n by ground recognition. It remains false in N by Theorem 4.6, using the trivial forcing as the first step.

If N recognizes a premouse P, its height is below Ω since Ω is preserved. Choose $Q \in \mathcal{C}_n^{M_n}$ above this height and fix an old code q for Q. The preceding argument gives $P \trianglelefteq Q$. Consequently P has an old real code, which satisfies the predicate in N by isomorphism invariance, and hence in M_n by the same absoluteness theorem. Conversely every old recognized code still passes the test by absoluteness. Thus the recognized structures agree, and their cofinal union is $M_n|\Omega$. □

Canonical trees. Write $M = M_n$ and use $<_M$ for its canonical construction order. The order on any initial segment is its own construction order, and every predecessor of an element of that segment belongs to it [20, 21]. All the following choices are made in M.

Lemma 4.10. *There is a canonical $\diamondsuit_\Omega$ sequence $\langle a_\lambda : \lambda < \Omega\rangle$ over M whose bounded computations agree between initial segments containing them.*

Proof. Set $a_\lambda = \varnothing$ at zero and successors. At a nonzero limit $\lambda < \Omega$, choose the $<_M$-least pair (X, C) such that $X \subseteq \lambda$, C is club in λ, and $a_\gamma \neq X \cap \gamma$ for all $\gamma \in C$. Put $a_\lambda = X$. There is a witness at every such λ. Choose a strictly increasing cofinal sequence $\langle \gamma_j : j < \omega\rangle$ in λ with $\gamma_j > j$. For $C = \{\gamma_j : j < \omega\}$, take $X \subseteq \omega$ with $j \in X$ iff $j \notin a_{\gamma_j}$. The witness test uses only the sequence already constructed and the indicated ordinal and sets. Thus any initial segment containing the recursion and the chosen witness sees the same least pair. In a bounded computation the least-pair clause requires a witness.

The verification that this recursion gives diamond is Jensen's condensation argument in M [21, Theorems 5.1 and 5.4(1)]. For the least counterexample (X, C) at Ω, use an ω-sound initial segment Q containing X, C and the sequence, with $\rho_\omega(Q) = \omega_1^Q = \Omega$. The projectum calculation for this choice is the one in [21, Sections 2 and 5]. New subsets of Ω appear cofinally below $M|\Omega^+$, giving sound levels projecting to at most Ω. A sound level containing Ω cannot project below Ω, since the hull of a smaller projectum and the standard parameters would be countable in M. Take a countable elementary hull of Q containing these data and the standard parameters up to a degree witnessing $\rho_\omega(Q) = \Omega$. If δ is its intersection with Ω, the inverse collapse has critical point and collapsed projectum both equal to δ. The condensation alternative involving an extender of length δ is excluded here. Its critical point $\nu < \delta$ is countable in Q. A surjection $s : \omega \to \nu$ in Q would have the same values in the degree-zero ultrapower, since every value is below the critical point, contrary to its range being $j(\nu)$. The collapse is therefore an initial segment of M.

The collapsed pair $(X \cap \delta, C \cap \delta)$ is the least counterexample to the sequence below δ in that segment. The bounded witness test and the initiality of the

construction order make it least in M as well. Since $\delta \in C$, the identity $a_\delta = X \cap \delta$ contradicts the counterexample. $\square$

Use this diamond in the simultaneous construction of Lemma 2.5, with the $<_M$-least choice at every step. It gives independent normal splitting Suslin trees $\vec{S}^n = \langle S^n_\xi : \xi < \Omega \rangle$ in M. In particular, the finite product argument there seals each guessed maximal antichain in the appropriate product. At each level use the $<_M$-least enumeration by an initial segment of ω to label its nodes. These labels will be used in the decoder.

4.3 The preparatory branch extension W_n

We now pass to the ground model over which the coding iteration will be performed. Throughout this subsection, ω_1 denotes $\omega_1^{M_n}$. Let

$$\vec{S}^n = \langle S^n_\xi \mid \xi < \omega_1 \rangle$$

be the independent sequence just constructed. All products in this subsection are computed in M_n.

Definition 4.11. *For $A \subseteq \omega_1$ with $A \in M_n$, put*

$$\mathrm{Br}^n_A := \prod_{\xi \in A}^{\mathrm{fs}} S^n_\xi.$$

We write

$$\mathrm{Br}_n := \mathrm{Br}^n_{\omega_1} = \prod_{\xi < \omega_1}^{\mathrm{fs}} S^n_\xi.$$

Let $H \subseteq \mathrm{Br}_n$ be M_n-generic. For $\xi < \omega_1$, define

$$b_\xi := \bigcup \{ p(\xi) : p \in H \text{ and } \xi \in \mathrm{dom}(p) \}.$$

For $A \subseteq \omega_1$ in M_n, let

$$H_A := H \cap \mathrm{Br}^n_A \,.$$

Finally, put

$$W_n := M_n[H].$$

Thus b_ξ is the branch added through S^n_ξ by the ξ-th coordinate. As before, to *destroy* a tree S^n_ξ means to add a cofinal generic branch through it.

Lemma 4.12. *Let $A, B \in M_n$ satisfy*

$$A \subseteq B \subseteq \omega_1.$$

Then the following hold in M_n.

1. Br^n_A *is c.c.c.*

2. Br^n_A *is a complete subforcing of* Br^n_B*, and*

$$\mathrm{Br}^n_B \cong \mathrm{Br}^n_A \times \mathrm{Br}^n_{B\setminus A}\,.$$

3. *If* $\xi \notin A$*, then*

$$\mathrm{Br}^n_A \Vdash \text{“}S^n_\xi \text{ is a Suslin tree.”}$$

Proof. The sequence $\vec{S}^n$ is independent in M_n, so every nonempty finite product of its members is a Suslin tree. Apply Lemma 2.2 inside M_n. The second clause is the coordinatewise factorization of the finite-support product. For the last clause, note that

$$\mathrm{Br}^n_A \times S^n_\xi \cong \mathrm{Br}^n_{A\cup\{\xi\}},$$

which is c.c.c. by the first clause. The product characterization of preservation of a Suslin tree therefore gives the conclusion. □

Theorem 4.13. *The model* $W_n = M_n[H]$ *has the following properties.*

1. *The forcing* Br_n *is c.c.c. and has size* $\aleph_1$ *in* M_n*. Consequently,* W_n *preserves every cardinal and cofinality of* M_n*, and*

$$W_n \models \mathsf{CH}.$$

2. *Every* S^n_ξ *is destroyed in* W_n*, as witnessed by the branch* b_ξ*.*

3. *If* $A \subseteq \omega_1$ *belongs to* M_n*, then* H_A *is* Br^n_A*-generic over* M_n*, and, for every* $\xi < \omega_1$*,*

$$M_n[H_A] \models \text{“}S^n_\xi \text{ has a cofinal branch”} \quad\Longleftrightarrow\quad \xi \in A.$$

4. *Every real in* W_n *belongs to* $M_n[H_A]$ *for some countable set* $A \subseteq \omega_1$ *with* $A \in M_n$*.*

5. *More generally, every member of* $H(\omega_1)^{W_n}$ *belongs to* $M_n[H_A]$ *for some countable set* $A \subseteq \omega_1$ *with* $A \in M_n$*.*

Proof. This is the argument of Theorem 2.7, carried out inside M_n, with $\vec{S}$, Br, and L replaced by $\vec{S}^n$, Br_n, and M_n, respectively. □

The last two clauses are the branch-coordinate localization needed by the coding argument. We record a form in which a prescribed countable set of coordinates may simultaneously be retained.

Corollary 4.14. *Let $a \in H(\omega_1)^{W_n}$, and let $C \in [\omega_1]^\omega \cap M_n$. Then there is $A \in [\omega_1]^\omega \cap M_n$ such that*

$$C \subseteq A, \qquad a \in M_n[H_A],$$

and

$$M_n[H_A] \models \text{“}S^n_\xi \text{ is a Suslin tree”} \qquad (\xi \in \omega_1 \setminus A).$$

In particular, no cofinal branch through any S^n_ξ with $\xi \notin A$ belongs to $M_n[H_A]$.

Proof. By Theorem 4.13(5), choose a countable branch support $A_0 \in M_n$ for a, and put

$$A = A_0 \cup C.$$

Then A is countable and belongs to M_n. Since $\mathrm{Br}^n_{A_0}$ is a complete subforcing of Br^n_A, we have $a \in M_n[H_A]$. The assertion about the unused trees follows from Lemma 4.12(3). □

4.4 The coding over M_n

Write $m^n_\infty = M_n|\Omega$ and use its fixed ordinal code, with all premouse predicates, in $L[m^n_\infty, r]$. This code is in M_n, and its transitive collapse recovers the lower part.

The almost disjoint family. Fix a recursive injection $\#\colon 2^{<\omega} \to \omega$. In M define, for $\xi < \Omega$,

$$x_\xi = \min_{<_M}\bigl(2^\omega \setminus \{x_\zeta : \zeta < \xi\}\bigr), \qquad d_\xi = \{\#(x_\xi \restriction k) : k < \omega\}.$$

Here $<_M$ is the canonical construction order on M. Each step is defined because ξ is countable and $M \models \mathsf{CH}$. The set d_ξ is infinite, and if $\zeta < \xi$ and k is the first position at which x_ζ and x_ξ differ, then

$$d_\zeta \cap d_\xi = \{\#(x_\xi \restriction j) : j \leq k\}.$$

Thus $D^n = \langle d_\xi : \xi < \Omega\rangle$ is almost disjoint. Write $D = D^n$ throughout this subsection. Its initial segments are computed in initial segments of M below Ω, as shown below.

For $Y \subseteq \Omega$, the conditions of $\mathbb{A}^n(Y)$ are pairs $(s, F) \in [\omega]^{<\omega} \times [Y]^{<\omega}$, ordered by

$$(t, G) \leq (s, F) \quad\Longleftrightarrow\quad s \subseteq t, \quad F \subseteq G, \quad (t \setminus s) \cap \bigcup_{\xi \in F} d_\xi = \varnothing.$$

Conditions with a common stem are centered. For the union r of the generic stems, $\xi \in Y$ implies $r \cap d_\xi$ is finite by the density of conditions with $\xi \in F$. For $\xi \notin Y$ one can add arbitrarily many points of d_ξ to the stem, since each intersection $d_\xi \cap d_\zeta$ for $\zeta \in F$ is finite. Thus $Y = \{\xi < \Omega : |r \cap d_\xi| < \aleph_0\}$.

The coding sets. Fix in M a bijection $\varrho_n\colon ({}^{<\Omega}2)^M \to \Omega$, using CH in M. Let $\mathbb{C}_{M_n} = ({}^{<\Omega}2)^M$, ordered by end-extension. This is the dense presentation by initial segments of $\mathrm{Add}(\omega_1, 1)^M$. For a $\mathbb{C}_{M_n}$-generic g over the current model set

$$h(g) = \{\varrho_n(g \restriction \eta) : \eta < \Omega\},$$
$$\mathrm{Sel}(h, u) = \{\tau_u(\gamma, k) : \gamma \in h,\ k < \omega\}, \qquad \tau_u(\gamma, k) = \omega\gamma + 2k + u(k).$$

We call $h(g)$ the *coding set* of g. Its cardinality is Ω. For distinct g and g' the intersection $h(g) \cap h(g')$ is bounded in Ω, since it consists of the images of their common initial segments. The decoding uses the set h itself and not ϱ_n.

To code $u \in 2^\omega$, let g be the generic real at a fresh Cohen coordinate and put $h = h(g)$. The forcing at the second step is $\mathbb{A}^n(Y_n(u, g))$, where $Y_n(u, g) \subseteq \Omega$ is defined below. We give the localization explicitly. It codes three objects: u, h, and the branches with indices in $\mathrm{Sel}(h, u)$.

Codes and restriction to countable heights. Fix a finite fragment of ZFC^- containing the instances of separation, collection, replacement, set-structure satisfaction and wellorder collapse used below, and add the axiom that ω_1 exists. A *decoding model* is a countable transitive model of this theory. Let $\langle\cdot,\cdot\rangle_{\mathrm{ord}}$ be Gödel's pairing function on ordinals, which enumerates pairs by increasing maximum and lexicographically for a fixed maximum; for triples iterate it. If N is a decoding model and $\alpha = \omega_1^N$, then pairing and its inverse commute with restriction to α. Indeed, the part of the enumeration through any pair of ordinals below α is countable in N. Conversely the rank of a pair in the enumeration is at least the maximum of its entries. The same restriction property holds for $\zeta \mapsto j \cdot \zeta + i$ with $i < j < \omega$. Also $\tau_u(\gamma, k) < \alpha$ for $\gamma < \alpha$ and $k < \omega$, since $\alpha = \omega_1^N$.

Present the nodes of each S^n_ξ by pairs (λ, i), where λ is the level and i is the index in the $<_M$-least bijection from an initial segment of ω onto that level. The bijection is chosen as soon as the level has been constructed. We use these relabelled trees throughout, with the tree orders transported along the bijections. In the codes below a node is identified with the ordinal pairing of λ, i. Thus the restriction of the indexed tree to level $< \alpha$ is obtained by keeping exactly the node indices below α. We regard each b_ξ as the set of indices of the nodes of its downward closure.

Localization. For $\gamma \in h$ and $k < \omega$ use the index $\tau_u(\gamma, k)$ and put

$$B = \{\langle\gamma, k, \zeta\rangle_{\mathrm{ord}} : \gamma \in h,\ k < \omega,\ \zeta \in b_{\tau_u(\gamma,k)}\},$$
$$X = \{3 \cdot k : k < \omega,\ u(k) = 1\} \cup \{3 \cdot \gamma + 1 : \gamma \in h\} \cup \{3 \cdot z + 2 : z \in B\}.$$

The odd part of $Y_n(u, g)$ codes X. For the even part, define a continuous increasing sequence in $\mathrm{acc}(h)$ by

$$c_0 = \min\{\zeta \in \mathrm{acc}(h) : \omega < \zeta\},$$
$$c_{\nu+1} = \min\{\zeta \in \mathrm{acc}(h) : c_\nu + \omega < \zeta\}, \quad c_\lambda = \sup_{\nu<\lambda} c_\nu \quad (\lambda \text{ a nonzero limit}).$$

Fix in M a family $\langle w_\eta : \omega \leq \eta < \Omega\rangle$ with w_η coding a wellorder of ω of type η. Define

$$E = w_{c_0} \cup \{c_\nu + k : \nu < \Omega,\ k \in w_{c_{\nu+1}}\},$$
$$Y_n(u, g) = \{2 \cdot \zeta : \zeta \in E\} \cup \{2 \cdot \zeta + 1 : \zeta \in X\}.$$

Here the wellorder codes use the fixed recursive coding of relations on ω. As in the construction over L, the even part localizes the code. The odd part gives the branch data. Apart from data in M, it uses only g, u and the selected branches.

Suppose N is a decoding model, with $\alpha = \omega_1^N$, and $Y_n(u, g) \cap \alpha \in N$. Then α belongs to $C = \{c_\nu : \nu < \Omega\}$, which is club because the sequence is continuous, strictly increasing and has length Ω. If $\alpha < c_0$, N reads w_{c_0} from the initial ω positions of the even part. If $c_\nu < \alpha < c_{\nu+1}$, then $c_\nu + \omega < \alpha$ by countability of c_ν in N. Using c_ν as an ordinal parameter, N reads $w_{c_{\nu+1}}$ from that block. In either case the relation read in N is well-founded in the ambient model, hence in N. Its collapse in N is an isomorphism onto an ordinal below α, since its domain is ω. The same map is an isomorphism in the ambient model, contradicting the prescribed order type. These cases exhaust the complement of C, since C is club. It follows that $h \cap \alpha$ is unbounded in α. The odd part yields $u, h \cap \alpha$ and, for $\gamma \in h \cap \alpha$, $k < \omega$, the branch $b_{\tau_u(\gamma,k)}$ restricted to levels below α. The restriction has one node at every such level, by the choice of node indices.

Correct local sequences. The set $\mathcal{I}_n$ is given by Definition 4.8. By Lemma 4.9, its recognized structures are unchanged in the small extensions used here. For each $\alpha < \Omega$ the intended local sequences are

$$D^\alpha = D \restriction \alpha, \qquad \vec{S}^{n,\alpha} = \langle S^n_\xi \restriction \alpha : \xi < \alpha\rangle,$$

where the latter restriction is by levels. A triple $(m, D', \vec{S}')$ consists of a nonempty well-founded premouse of height $\alpha = \mathrm{Ord}(m)$ which is the union of its proper initial segments, together with sequences of the indicated shape. The triple is *correct* if for every $\tau < \alpha$ there is $\beta \in (\tau, \alpha)$ such that $m|\beta \in \mathcal{I}_n$ and a computation in $m|\beta$ agrees with $D', \vec{S}'$ below τ. This means agreement on the entries of D' of index $< \tau$ and on the trees of index $< \tau$ restricted to levels $< \tau$, including node indices and orders. A *computation* is an element of $m|\beta$ recording an initial part of the constructions of D and $\vec{S}^n$.

We show that correct triples agree with M, and that they exist at club many heights. The canonical order of M restricts to the order of each initial

segment, and every $<_M$-predecessor of an element of an initial segment belongs to it. A computation of D records the sequence of the x_ξ's and the sets of their finite initial segments. The checks at each entry are

$$x_\xi \in 2^\omega, \qquad x_\xi \notin \{x_\zeta : \zeta < \xi\}, \qquad d_\xi = \{\#(x_\xi \restriction j) : j < \omega\},$$
$$(\forall y \in 2^\omega)(y <_M x_\xi \implies \exists \zeta < \xi \ (y = x_\zeta)).$$

The quantifier over y ranges over the initial segment containing the computation. All earlier possible choices belong to that segment, so induction on ξ shows agreement with the construction in M.

For the diamond sequence the computation records the witnesses to the least-pair recursion of Lemma 4.10. A bounded computation at a nonzero limit stage includes the pair, with leastness checked in its segment. The test is absolute once the earlier values are fixed, so the same induction applies.

For the construction of the trees, the computation records the tree orders below the current level, the antichains coded by the diamond guesses, the families of branches chosen at limit levels, and the level bijections. If the guess at a limit λ codes a maximal antichain A in the product of the trees with indices in a finite set e, restricted to levels below λ, the recorded family $\langle b_{\xi,j} : \xi \in e, \ j < \omega \rangle$ must seal A:

$$\forall f \in {}^e\omega \ \exists t \in A \ \forall \xi \in e \ (t(\xi) \in b_{\xi,f(\xi)}).$$

Here maximality refers to this countable product. In addition every node below level λ of each tree must lie on one of the recorded branches. Such families exist by the countable product argument of Lemma 2.5. The conditions on the choices only refer to the data constructed so far. Hence any two initial segments of M containing these data and a witness make the same least choice, since all earlier possible choices lie in the smaller segment. An initial segment of the sequences together with its computation is hereditarily countable in M, hence belongs to $M|\Omega = H^M_\Omega$ by the fine structure of the hierarchy [21, Section 2]. By Lemma 4.9 it belongs to a recognized segment below Ω. Consequently we may fix in M a function f on Ω such that $f(\tau) > \tau$ is the height of an initial segment in $\mathcal{I}_n$ containing the computations up to τ.

Correctness is a parameter-free Π^1_{n+1} property: well-foundedness of the codes is Π^1_1, the conditions on computations inside the coded structures are arithmetic, and the quantifiers over initial segments and computations range over the countable codes. The definition of $\mathcal{I}_n$ and the agreement of computations give

$$(m, D', \vec{S}') = (M|\alpha, D^\alpha, \vec{S}^{n,\alpha})$$

for every correct triple. To see that correct triples exist at club many heights, fix a sufficiently large H_Θ in the current small extension. Take countable $A \prec H_\Theta$ containing the parameters under consideration, m^n_∞, Ω,

the sequences $D, \vec{S}^n$, f, the map $\lambda \mapsto M|\lambda$, and ground enumerations of each $M|\lambda$, $0 < \lambda < \Omega$. For $\alpha = A \cap \Omega$, the enumerations and the level map give $A \cap m^n_\infty = M|\alpha$. The collapse N of A has $\omega_1^N = \alpha$, and sends $D, \vec{S}^n$ to $D^\alpha, \vec{S}^{n,\alpha}$. For $\xi < \alpha$, the node indices of S^n_ξ that are in A are exactly those below α, namely the indices of nodes at levels below α. These indices and the order between them are preserved by the collapse. Closure under f provides the required computations below α. The heights of a continuous chain of such submodels form a club.

Lemma 4.15 (Decoding)**.** *There are parameter-free formulas*

$$\Psi_n(r,u) \in \Pi^1_{n+2}, \qquad \Phi_n(u) \iff \exists r\, \Psi_n(r,u), \qquad \Phi_n \in \Sigma^1_{n+3},$$

with the following properties.

1. *If r is the real added at a coding stage for u, then $\Psi_n(r,u)$ holds in the resulting extension and in every further small extension.*

2. *If $\Psi_n(r,u)$ holds in a small extension, then $u \in L[m^n_\infty, r]$, and there is an unbounded $h \subseteq \Omega$ in $L[m^n_\infty, r]$ such that $L[m^n_\infty, r]$ contains a cofinal branch through S^n_ξ for every $\xi \in \mathrm{Sel}(h,u)$.*

Proof. Let $\mathrm{Dec}_n(c; r, u)$ assert that c codes a decoding model N containing r, u and a triple $(m, D', \vec{S}')$ correct in the ambient model, with $\mathrm{Ord}(m) = \omega_1^N = \alpha$. In N put

$$Y^N_r = \{\xi < \alpha : |r \cap d'_\xi| < \omega\}.$$

Read (u', h', B') from its odd part and set $B'_{\gamma,k} = \{\zeta : \langle \gamma, k, \zeta\rangle_{\mathrm{ord}} \in B'\}$. Let $\mathrm{Br}_\alpha(b, S)$ mean that $b \subseteq S$ is linearly ordered by the tree order and meets every level below α exactly once. The local test is

$$\mathrm{Test}(c; r, u) \iff u' = u \wedge \sup h' = \alpha \wedge \forall \gamma \in h'\, \forall k < \omega\ \mathrm{Br}_\alpha(B'_{\gamma,k}, S'_{\tau_u(\gamma,k)}).$$

The code includes satisfaction relations checked arithmetically by the Tarski clauses. Correctness gives $\mathrm{Dec}_n \in \Pi^1_{n+1}$, and Test is arithmetic in the code. Hence

$$\Psi_n(r,u) \iff \forall c\, (\mathrm{Dec}_n(c; r, u) \implies \mathrm{Test}(c; r, u))$$

is parameter-free Π^1_{n+2}.

For r added by $\mathbb{A}^n(Y_n(u,g))$, possibly restricted below a condition, put $Y_r = \{\xi < \Omega : |r \cap d_\xi| < \omega\}$. Correctness and localization give, for every decoding model for (r,u),

$$Y_r = Y_n(u,g), \qquad Y^N_r = Y_r \cap \alpha,$$
$$(u', h', B') = (u, h \cap \alpha, B \cap \alpha), \qquad \alpha \in \mathrm{acc}(h).$$

For each $\gamma \in h \cap \alpha$, $k < \omega$, the section $B'_{\gamma,k}$ is $b_{\tau_u(\gamma,k)} \restriction \alpha$. Thus the test holds. Theorem 4.6 gives persistence.

Conversely, suppose $\Psi_n(r,u)$. In $L[m_\infty^n, r]$, recover $D, \vec{S}^n$ by taking the union of the computations in m_∞^n. They agree by induction on the least choices, and their bounds are cofinal in Ω. Read $(\hat{u}, \hat{h}, \hat{B})$ from Y_r. At every height α of a decoding model for (r,u), correctness and restriction of the ordinal coding give

$$Y_r^N = Y_r \cap \alpha, \qquad (u', h', B') = (\hat{u}, \hat{h} \cap \alpha, \hat{B} \cap \alpha).$$

Such heights contain a club, so the tests give $\hat{u} = u$ and $\sup \hat{h} = \Omega$. For each $\gamma \in \hat{h}$, $k < \omega$, and $\xi = \tau_u(\gamma, k)$, those tests at heights $\alpha > \gamma, \xi$ give

$$\mathrm{Br}_\alpha(\hat{B}_{\gamma,k} \cap \alpha, S_\xi^n \restriction \alpha).$$

Any two nodes and any prescribed level are covered at such a height. Thus $\mathrm{Br}_\Omega(\hat{B}_{\gamma,k}, S_\xi^n)$, with $\hat{B}_{\gamma,k} \in L[m_\infty^n, r]$. □

Coding iterations and exactness

Put $\Omega = \omega_1^{M_n}$. For a set $E \in M_n$ of at most Ω Cohen indices, put

$$C_E^n = \mathrm{Fn}_{<\Omega}(E \times \Omega, 2)^{M_n}.$$

Use the same sets E_i of Cohen indices as in the construction over L, with the reindexing maps now taken in M_n. For $\beta \leq \Omega$, the forcing through the stages $i < \beta$ is

$$P_\beta^n = C_{E_{<\beta}}^n * \dot{R}_\beta^n, \qquad E_{<\beta} = \bigcup_{i<\beta} E_i, \qquad R_\beta^n = \operatorname*{fs}_{i<\beta} * \dot{A}_i.$$

At stage i, names over P_i^n specify whether to code a real, the real u_i to be coded, the selected coordinate in E_i, and the ground-model isomorphism onto a restriction of $C_{E_i}^n$. If the stage codes a real, let g_i be the Cohen subset of Ω obtained at the selected coordinate after applying this isomorphism. Then $\dot{A}_i$ is a name for $\mathbb{A}^n(Y_n(u_i, g_i))$, possibly restricted below a condition named over $P_i^n * C_{E_i}^n$. If there is no coding at stage i, its second factor is trivial.

The proof of Lemma 2.9 applies with M_n in place of L. The forcing C_E^n splits over disjoint index sets in M_n. Given u_i and g_i, the lower part and the indexed family of selected branches determine Y_n in any model containing these data and M_n. The almost disjoint order on the finite pairs is determined by D^n, Y_n, and the condition below which the forcing is restricted. The isomorphisms used to restrict the Cohen forcing change only a bounded part of each coordinate. Composing these isomorphisms with the ones already specified by the iteration leaves the selected g_i unchanged. This gives the projections to initial segments, the presentations below conditions, and the extension of a tail defined below a Boolean value by using trivial second factors off that value.

For a product of two tails over the same initial segment, use disjoint sets of Cohen indices and translate the names of each factor by the complete embeddings of its initial segments. Either factor can be taken first, since the translated names give the same g_i, u_i, and Y_n in both orders. If a product $R \times Q_j$ is already given and T is a continuation of Q_j below a condition, retain the reindexing of both factors and extend the reindexing on the Q_j side to the new indices for T. Use the image of the same Boolean value to determine where the almost disjoint factors of T are used. The embeddings of the prescribed initial segments are unchanged, including at limit stages by finite support.

Every such iteration of length at most Ω, including the preparatory branch product, has a dense suborder of size Ω, preserves cardinals, and forces CH. Indeed the Cohen product can be taken before the branch forcing. It is countably closed and has size Ω in M_n. In the resulting extension the branch product is c.c.c. by the same countable-closure and finite-product argument as over L. Over W_n the Cohen product adds no countable sequences of ordinals by Easton's lemma. Each almost disjoint factor is Knaster, including below a condition. The finite-support iteration is therefore c.c.c. and has a dense suborder of size at most Ω consisting of finite functions whose values are almost disjoint conditions. The same argument applies to the product presentations and to tails extended off a Boolean value as above. In particular the complete embeddings between initial segments give small two-step extensions of M_n. To see the size bound for the two-step forcing, pass to dense suborders and use mixed names for the conditions in an enumeration of the dense suborder of the quotient.

Every real in an extension by an iteration of length Ω belongs to the extension by some P^n_β, $\beta < \Omega$, as in Lemma 2.10. After forcing with the full Cohen product, take a nice R^n_Ω-name for the real in the dense suborder of finite functions. Its countable ordinal code belongs to W_n by Easton's lemma. Choose β above all the indices in the supports of conditions in the name. The factors before β depend only on the Cohen coordinates before β and their own initial segments. Thus the dense suborder of R^n_β is the same after forcing with either $C^n_{E_{<\beta}}$ or the full Cohen product. Evaluation of the same name gives the real in the P^n_β-extension.

Theorem 4.16. *In a generic extension of W_n by a coding iteration of length at most Ω,*

$$\Phi_n(u) \quad \Longleftrightarrow \quad \exists i \ (i \text{ is a coding stage and } u_i = u).$$

Proof. At a coding stage the almost disjoint generic real codes $Y_n(u_i, g_i)$ also when the factor is restricted below a condition. The Cohen subset g_i is obtained from the specified ground-model isomorphism, so Lemma 4.15 applies and gives the same witness in the final extension.

Conversely, suppose r witnesses $\Phi_n(u)$ and no stage codes u. Fix the generic K for the full Cohen product. In $W_n[K]$ take nice names for r, u using finite functions as conditions. Let J be the countable set of stages obtained by closing their supports under the supports of the names used at those stages. These names specify whether to code a real, which real to code, the selected Cohen coordinate and the associated isomorphism, and any condition below which the almost disjoint factor is restricted. The finite-support iteration on J embeds completely into the full iteration. At a successor in J the name for the factor uses only the earlier stages in J, and at limits the embeddings commute with the finite-support projections. The countable ordinal code for these names belongs to W_n. By the c.c.c. of the branch product it has a name supported by a countable $B \subseteq \Omega$ in M_n.

By Lemma 4.15 there is an unbounded set h_r such that, for every $\gamma \in h_r$ and $k < \omega$, $L[r, m_\infty^n]$ contains a cofinal branch through $S^n_{\tau_u(\gamma,k)}$. The selected Cohen subsets at different coding stages in J disagree on an unbounded set, since the original coordinates do and the ground-model isomorphisms change them only on bounded sets. Hence their associated sets h_i have bounded pairwise intersections. Choose $\gamma \in h_r$ above the bounds on those intersections and above B. There is at most one coding stage $i \in J$ such that $\gamma \in h_i$. If there is one, then $u_i \neq u$, so choose k with $u_i(k) \neq u(k)$. Otherwise choose any $k < \omega$. Then

$$\xi = \tau_u(\gamma, k) \notin B \cup \bigcup_{\substack{i \in J \\ i \text{ is a coding stage}}} \mathrm{Sel}(h_i, u_i).$$

Work over $M_n[K][H \restriction (\Omega \setminus \{\xi\})]$. The ordinal code for the names on J belongs to this model. As in Lemma 2.11, interpret the names on J and use trivial factors from the first stage requiring the missing branch at ξ onward. At each earlier stage all data for Y_n are present in the model. The resulting iteration has the same dense suborder of finite functions in this model and in $W_n[K]$. This follows by induction on the stages, using absoluteness of membership and order for the next factor in common generic extensions by the preceding stages. At a limit each finite support is bounded. The modified iteration embeds completely into the iteration on J, and the embedding identifies the restrictions to the Boolean values on which the missing branch is never required. At successors the factors agree below these values. At a limit the value is the meet of the earlier ones. In the given generic extension, no stage on J requires the branch at ξ. The induced generic for the modified iteration thus gives an extension of $M_n[K][H \restriction (\Omega \setminus \{\xi\})]$ containing r and u.

The tree S^n_ξ is Suslin in $M_n[K][H \restriction (\Omega \setminus \{\xi\})]$, since the ground-model Cohen product can be taken before the branch forcing. The product of the modified iteration with S^n_ξ is c.c.c. by the Knaster property of the factors and finite support. Thus S^n_ξ is still Suslin in the extension containing r. This extension also contains m_∞^n, hence $L[r, m_\infty^n]$ and a cofinal branch through

S^n_ξ, a contradiction. □

4.5 Transfer of the uniformization construction

Fix a recursive universal enumeration of Π^1_{n+3} formulas with two free real variables, and write A^n_m for the corresponding relations. Apply Theorem 3.1 with W_n, A^n_m, and Φ_n in place of W, A_m, and Φ.

The preceding arguments give the complete embeddings for initial segments and products, the presentations below conditions, and the extension of products preserving specified initial segments. The size bound gives a set of choices of names for choice functions and tails in W_n. The two quotient generics in a product are mutually generic over the common initial extension. The argument by generic intersection therefore applies both to the choice functions and to the restrictions on coding reals absent from the common extension. The definition of allowability uses the forcing relation for A^n_m with respect to the same embeddings.

For persistence, write

$$(x, y) \notin A^n_m \quad \iff \quad \exists z\, \theta_m(z, x, y), \qquad \theta_m \in \Pi^1_{n+2}.$$

If the extension by one factor supplies a witness z, the same z witnesses nonmembership in the product extension by Theorem 4.6. The same applies when a tail forcing nonmembership is appended to one factor in the proof of preservation after further restrictions on coding. If a selected value fails to belong to A^n_m in the final extension, a witness z belongs to the extension by a bounded initial segment, and the same theorem reflects $\theta_m(z, x, y)$ to that extension. All these inclusions are small two-step extensions by the preceding size bound. Thus the arguments for persistence, the least rank and its minimizers, the first initial segment containing a new minimizer, and the preservation of the selected value apply over M_n.

Theorem 4.17. *Let $1 \leq n < \omega$ and suppose that $M^\#_n$ exists. There is a cardinal-preserving set-generic extension of M_n satisfying* CH *and the Π^1_{n+3}-uniformization property.*

Proof. First force over M_n with the branch product producing W_n, then use the uniformization construction with the hypotheses just verified. The resulting coding iteration has length Ω. In the final extension put

$$U^n_m(x, y) \quad \iff \quad (x, y) \in A^n_m \ \wedge \ \neg\Phi_n(\langle x, y, m\rangle).$$

The construction selects exactly one value in each nonempty section, and this is the unique value in the section for which $\langle x, y, m\rangle$ is never coded. By Theorem 4.16, U^n_m is the graph of the resulting uniformizing function. Lemma 4.15 gives $\Phi_n \in \Sigma^1_{n+3}$, so $U^n_m \in \Pi^1_{n+3}$. Real parameters can be included in the first argument using the fixed recursive pairing function. The cardinal and continuum assertions follow from the preservation argument above. □

5 Further possible applications and open problems

We close with two directions suggested by the construction. First, the method should have an analogue in the generalized Baire-space context at a regular uncountable cardinal κ. In particular, one may ask whether the Π^1_1-uniformization problem for $\kappa^{<\kappa}$ can consistently have a positive solution.

The methods in this paper leave the following question open:

Question 1. *Can the Π^1_n-uniformization property be forced over L for $n > 3$?*

The method introduced here is so far limited to local effects. It would be interesting to force a less local, or even global, behaviour:

Question 2. *Given a pair $n, m \in \omega$ such that $n \neq m$ and $n \neq m + 1$, can one force a universe in which the Π^1_n- and the Π^1_m-uniformization properties both hold?*

Finally, it seems worthwhile to combine the present method with other forcing constructions in order to obtain models with properties that are usually consequences of determinacy assumptions. We record one paradigmatic question:

Question 3. *Assume some sufficiently big large cardinal notion which can exist in L. Is there a model which satisfies the Π^1_3-uniformization property and in which every projective set of reals is Lebesgue measurable?*

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